

4

Imaging in Resist: Standing Waves and Swing Curves

The previous chapter developed a comprehensive theory for calculating aerial images, the image of a photomask as projected by the imaging tool into air. But of course the real goal is to project that image into photoresist, where chemical reactions responding to the local exposure dose will transform the intensity image into a latent image of exposed and unexposed material. The coupling of the aerial image into the photoresist involves the behavior of light as it encounters interfaces between materials – reflection and transmission – and the propagation of light through an absorbing medium.

One of the unique aspects of the image in resist (as opposed to the aerial image) comes from reflections off the substrate (or films coated on the substrate). The total energy in the resist is the sum of the downward-propagating image and the upward-propagating reflected image. Since these two images will interfere, the results are highly dependent on the thickness of the resist, and possibly other films above or below the resist. Two consequences of this interference are (1) standing waves – a sinusoidal variation of dose through the thickness of the resist – and (2) swing curves – a sinusoidal dependence of the energy coupled into the resist as a function of resist thickness.

A thorough understanding of these phenomena, plus a recognition of their detrimental impact on linewidth control, leads to a search for methods to reduce standing waves and swing curves. A description of antireflection coatings, both top and bottom, will show how linewidth control can be reclaimed through effective reflection control.

Finally, examining how light propagates through the photoresist, including all of the various thin film interference effects, allows for a rigorous understanding of the difference between an aerial image and an image in resist.

4.1 Standing Waves

Standing waves occur whenever two waves, traveling in opposite directions and with a fixed phase relationship to each other, combine. Examples of standing waves abound, from vibrational waves traveling down a guitar string and reflecting off the fixed end of that string, to acoustic waves bouncing within the metal tube of a pipe organ, to the design of impedance matching connections for coaxial television cables. In lithography, light passing through the photoresist is reflected off the substrate. This reflected light wave interferes with the light wave traveling down to produce a standing wave pattern.^{1,2}

4.1.1 The Nature of Standing Waves

Before looking at the lithographic case in great detail, we'll examine the basic nature of standing waves by considering an exceedingly simplified version of light standing waves. Suppose a plane wave of monochromatic light traveling through air (which we'll assume to be nonabsorbing with an index of refraction of 1.0) is normally incident on a mirror. What is the total light intensity in the vicinity of the mirror? The total electric field will be the sum of the wave traveling toward the mirror and the reflected wave. Letting z be the direction the incident light is traveling in, the electric field E_i of a plane wave traveling in this direction has the mathematical form

$$E_i(z) = Ae^{ikz} \quad (4.1)$$

where A is the amplitude of the plane wave and k is the propagation constant (equal to $2\pi n/\lambda$ where n is the refractive index of the medium and λ is the vacuum wavelength of the light). (This equation requires an assumption about the sign convention for the phasor representation of a sinusoidal wave, as discussed in Chapter 2. Unfortunately, there is no standard and the convention commonly used for thin film calculations is the opposite to that normally used for imaging. In this book, a plane wave traveling in the $+z$ -direction is represented by $\exp(+ikz)$ and the imaginary part of the index of refraction must be positive to represent an absorbing media.) For simplicity, let $z = 0$ define the position of the mirror. Letting ρ be the electric field reflectivity of the mirror, the reflected light electric field will be

$$E_R(z) = \rho E_i(0)e^{-ikz} = \rho Ae^{-ikz} \quad (4.2)$$

The total electric field is just the sum of the incident and reflected fields.

$$E_T(z) = E_R(z) + E_i(z) = A(e^{+ikz} + \rho e^{-ikz}) \quad (4.3)$$

The special case of a perfectly reflective mirror ($\rho = -1$) provides an especially simple result. The two complex exponentials in Equation (4.3) become a sinusoid by using the Euler identity.

$$E_T(z) = A(e^{+ikz} - e^{-ikz}) = i2A \sin(kz) \quad (4.4)$$

The intensity of the light in this region is the 'interference pattern' between the incident and reflected beams:

$$I_T(z) = |E_T(z)|^2 = 4A^2 \sin^2(kz) = 2A^2(1 - \cos(2kz)) \quad (4.5)$$

The most striking aspect of this simple result is that the traveling wave has been converted to a standing wave. The plane wave in Equation (4.1) is ‘traveling’, meaning that the phase of the electric field varies linearly with distance traveled. Literally, the wave moves through its ups and downs of the sinusoidal phase cycle as it travels, while the amplitude remains constant. In contrast, the wave in Equation (4.4) is stationary; it is ‘standing’ in place. The amplitude of the electric field varies sinusoidally with position z , but the phase remains fixed and independent of position.

(As an interesting historical footnote, the mathematical experiment described above is almost identical to a famous and seminal experiment performed in 1888 by Heinrich Hertz. Attempting to verify Maxwell’s new theory of electromagnetic radiation, Hertz covered one of the walls of his laboratory with copper and reflected radio waves off this ‘mirror’. Using a wire loop detector sensitive to the magnitude of the electric field, he measured the resulting sinusoidal standing wave, which agreed well with Maxwell’s theory. Otto Wiener first demonstrated that light exhibits standing waves in 1890 by exposing a photographic film tilted at a small angle to a normally illuminated plane mirror.)

4.1.2 Standing Waves for Normally Incident Light in a Single Film

The very simple standing wave result derived above shows us the basic form of standing waves caused by plane waves traveling in opposite directions. For the case of standing waves in a photoresist, the simplest example is a thin layer of photoresist coated on a thick (semi-infinite) substrate. Let us begin with the simple geometry shown in Figure 4.1a. A thin photoresist (layer 2) rests on a thick substrate (layer 3) in air (layer 1). Each material has optical properties governed by its complex index of refraction, $\mathbf{n} = n + i\kappa$, where n is the real index of refraction and κ is the imaginary part, sometimes called the extinction coefficient. This latter name comes from the relationship between the imaginary part of the refractive index and α , the absorption coefficient of the material (as derived in Chapter 2).

$$\alpha = \frac{4\pi\kappa}{\lambda} \quad (4.6)$$

Consider now the propagation of light through this film stack. We will begin with illumination of the stack by a monochromatic plane wave normally incident on the resist. When

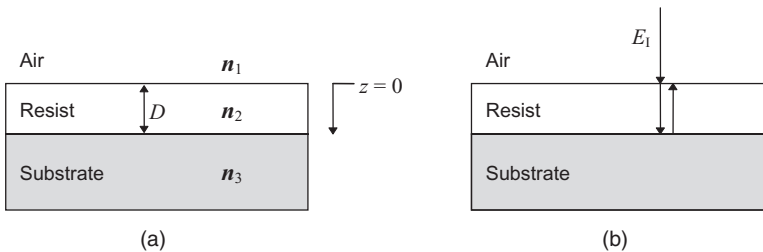


Figure 4.1 Film stack showing (a) the geometry for the standing wave derivation, and (b) a normally incident electric field E_1

this plane wave strikes the resist surface, some of the light will be transmitted and some will be reflected. The amount of each is determined by the transmission and reflection coefficients. Defined as the ratio of the transmitted to incident electric field, the *transmission coefficient* τ_{ij} for a normally incident plane wave transmitting from layer i to layer j is given by

$$\tau_{ij} = \frac{2n_i}{n_i + n_j} \quad (4.7)$$

In general, the transmission coefficient will be complex, indicating that when light is transmitted from one material to another, both the magnitude and the phase of the electric field will change. Similarly, the light reflected off layer j back into layer i is given by the *reflection coefficient* ρ_{ij} :

$$\rho_{ij} = \frac{n_i - n_j}{n_i + n_j} \quad (4.8)$$

Both of these equations are derived from the requirement that electric and magnetic field components that are within (or tangential to) the plane of the interface between materials i and j must be continuous across that boundary.

If an electric field E_i is incident on the photoresist (see Figure 4.1b), the transmitted electric field will be $\tau_{12}E_i$. The transmitted plane wave will now travel down through the photoresist. As it travels, the wave will change phase sinusoidally with distance and undergo absorption. Both of these effects are given by the standard description of a plane wave as a complex exponential:

$$E(z) = e^{ikz} = e^{i2\pi n_2 z / \lambda} \quad (4.9)$$

where this $E(z)$ represents a plane wave traveling in the $+z$ -direction. Using the coordinates defined in Figure 4.1a, the transmitted electric field propagating through the resist, E_0 , will be given by

$$E_0(z) = \tau_{12}E_i e^{i2\pi n_2 z / \lambda} \quad (4.10)$$

Eventually, the wave will travel through the resist thickness D and strike the substrate, where it will be partially reflected. The reflected wave, E_1 , just after reflection will be

$$E_1(z = D) = \tau_{12}E_i \rho_{23} e^{i2\pi n_2 D / \lambda} \quad (4.11)$$

The complex exponential term in the above equation has some physical significance: it represents the electric field transmitted from the top to the bottom of the photoresist and is called the *internal transmittance* of the resist, τ_D .

$$\tau_D = e^{i2\pi n_2 D / \lambda} \quad (4.12)$$

As the reflected wave travels back up through the resist, the distance traveled will be $(D-z)$ and the propagation will be similar to that described by Equation (4.9):

$$E_1(z) = \tau_{12}E_i \rho_{23} \tau_D e^{i2\pi n_2 (D-z) / \lambda} = \tau_{12}E_i \rho_{23} \tau_D^2 e^{-i2\pi n_2 z / \lambda} \quad (4.13)$$

So far, our incident wave (E_1) has been transmitted in the photoresist (E_0) and then reflected off the substrate (E_1), as pictured in Figure 4.1b. The resultant electric field in the photoresist will be the sum of E_0 and E_1 .

$$E_0(z) + E_1(z) = \tau_{12}E_1(e^{i2\pi n_2 z/\lambda} + \rho_{23}\tau_D^2 e^{-i2\pi n_2 z/\lambda}) \quad (4.14)$$

This sum is not the total electric field in the photoresist. The wave E_1 will travel to the top of the resist where it will be reflected by the air–resist interface. The new wave, E_2 , will travel down through the resist where it will be reflected by the substrate, giving another wave E_3 . The next two waves are given by

$$\begin{aligned} E_2(z) &= E_1 \rho_{21} \rho_{23} \tau_{12} \tau_D^2 e^{ik_2 z} \\ E_3(z) &= E_1 \rho_{21} \rho_{23}^2 \tau_{12} \tau_D^4 e^{-ik_2 z} \end{aligned} \quad (4.15)$$

The total electric field within the thin film, $E_T(z)$, is the sum of each $E_j(z)$. Performing this summation gives

$$E_T(z) = E_1 \tau_{12} (e^{ik_2 z} + \rho_{23} \tau_D^2 e^{-ik_2 z}) S \quad (4.16)$$

where $S = 1 + \rho_{21} \rho_{23} \tau_D^2 (1 + \rho_{21} \rho_{23} \tau_D^2 (1 + \dots$

The summation S is simply a geometric series and converges to

$$S = \frac{1}{1 - \rho_{21} \rho_{23} \tau_D^2} = \frac{1}{1 + \rho_{12} \rho_{23} \tau_D^2} \quad (4.17)$$

Thus, the total electric field in the resist will be³

$$\frac{E_T(z)}{E_1} = \frac{\tau_{12}(e^{i2\pi n_2 z/\lambda} + \rho_{23} \tau_D^2 e^{-i2\pi n_2 z/\lambda})}{1 + \rho_{12} \rho_{23} \tau_D^2} \quad (4.18)$$

Although we have determined an expression for the standing wave electric field, it is the intensity of the light that causes exposure. Calculation of the relative intensity from Equation (4.18) leads to a somewhat messy result, but a few algebraic simplifications allow for a reasonably useful form:

$$\begin{aligned} I(z) &= \frac{n_2 |E(z)|^2}{n_1 |E_1|^2} \\ &= T_{\text{eff}} \left[(e^{-\alpha z} + |\rho_{23}|^2 e^{-\alpha(2D-z)}) + 2|\rho_{23}| e^{-\alpha D} \cos(4\pi n_2 (D-z)/\lambda + \phi_{23}) \right] \end{aligned} \quad (4.19)$$

where $\rho_{23} = |\rho_{23}| e^{i\phi_{23}}$, and

$$T_{\text{eff}} = \frac{\frac{n_2}{n_1} |\tau_{12}|^2}{|1 + \rho_{12} \rho_{23} \tau_D^2|^2} = \frac{T_{12}}{|1 + \rho_{12} \rho_{23} \tau_D^2|^2}, \text{ the effective transmittance into the photoresist.}$$

This equation is graphed in Figure 4.2 for an i-line photoresist with typical properties on a silicon substrate. By comparing the equation to the graph, many important aspects of the standing wave effect become apparent. The most striking feature of the standing wave

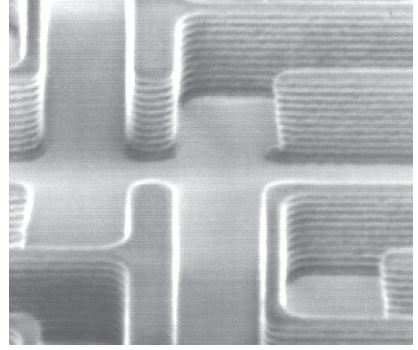
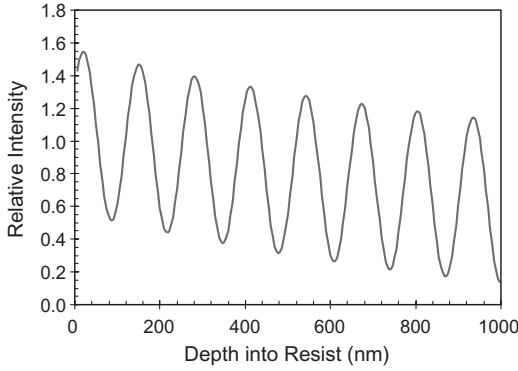


Figure 4.2 Standing wave intensity in one micron of photoresist on a silicon substrate for an *i*-line exposure

plot is its sinusoidal variation. The cosine term in Equation (4.19) shows that the period of the standing wave is given by

$$\text{Period} = \lambda/2n_2 \quad (4.20)$$

The amplitude of the standing waves is given by the multiplier of the cosine in Equation (4.19). It is quite apparent that there are two basic ways to reduce the amplitude of the standing wave intensity. The first is to reduce the reflectivity of the substrate (reduce ρ_{23}). The use of an antireflection coating on the substrate is one of the most common methods of reducing standing waves (see section 4.3). The second method for reducing the standing wave intensity that Equation (4.19) suggests is to increase absorption in the resist (reduce the $e^{-\alpha d}$ term). This is accomplished by adding a dye to the photoresist (increasing α). Finally, the ‘bulk’ intensity variation, the variation of the intensity averaged over the standing wave period, is given by the pair of exponential terms being added to the cosine in Equation (4.19). The bulk effect is a function of both absorption in the resist and the reflectivity of the substrate.

As an aside, it is worth mentioning one difficulty in using Equation (4.19): the conversion of the complex substrate reflectivity from rectangular to polar form. The calculation of the magnitude of the reflection coefficient is straightforward:

$$|\rho_{23}| = \sqrt{\frac{(n_2 - n_3)^2 + (\kappa_2 - \kappa_3)^2}{(n_2 + n_3)^2 + (\kappa_2 + \kappa_3)^2}} \quad (4.21)$$

However, determining the angle of the reflectivity using an arctangent function (as one might commonly do in spreadsheet calculations) can introduce an ambiguity if the range over which the arctangent is defined is not properly taken into account. Defining two simplifying variables x and y ,

$$\begin{aligned} x &= (n_2^2 + \kappa_2^2) - (n_3^2 + \kappa_3^2) \\ y &= 2(n_3\kappa_2 - n_2\kappa_3) \end{aligned} \quad (4.22)$$

(1) If $\tan^{-1}$ is defined on the interval $[0, \pi)$, then

$$\begin{aligned}\theta_{23} &= \tan^{-1}\left(\frac{y}{x}\right), & y > 0 \\ \theta_{23} &= \tan^{-1}\left(\frac{y}{x}\right) + \pi, & y < 0\end{aligned}\quad (4.23)$$

(2) If $\tan^{-1}$ is defined on the interval $[-\pi/2, \pi/2)$, then

$$\begin{aligned}\theta_{23} &= \tan^{-1}\left(\frac{y}{x}\right), & x > 0 \\ \theta_{23} &= \tan^{-1}\left(\frac{y}{x}\right) + \pi, & x < 0\end{aligned}\quad (4.24)$$

The cases of $y = 0$ or $x = 0$ can be handled separately from the arctangent evaluation.

4.1.3 Standing Waves in a Multiple-Layer Film Stack

It is very common to have more than one film coated on a substrate. An analysis similar to that for one film yields the following result for the electric field in the top layer of an m -1 layer system:

$$\frac{E_2(z)}{E_1} = \frac{\tau_{12}(e^{i2\pi n_2 z/\lambda} + \rho'_{23}\tau_D^2 e^{-i2\pi n_2 z/\lambda})}{1 + \rho_{12}\rho'_{23}\tau_D^2} \quad (4.25)$$

where

$$\begin{aligned}\rho'_{23} &= \frac{n_2 - n_3 X_3}{n_2 + n_3 X_3} \\ X_3 &= \frac{1 - \rho'_{34}\tau_{D3}^2}{1 + \rho'_{34}\tau_{D3}^2} \\ \rho'_{34} &= \frac{n_3 - n_4 X_4}{n_3 + n_4 X_4} \\ &\vdots \\ &\vdots \\ X_m &= \frac{1 - \rho_{m,m+1}\tau_{Dm}^2}{1 + \rho_{m,m+1}\tau_{Dm}^2} \\ \rho_{m,m+1} &= \frac{n_m - n_{m+1}}{n_m + n_{m+1}} \\ \tau_{Dj} &= e^{ik_j D_j}\end{aligned}$$

and all other parameters are as previously defined. The parameter ρ'_{23} is the effective reflection coefficient between the thin resist film and what lies beneath it. Once the effective reflection coefficient of the substrate film stack is determined, Equation (4.25) becomes identical to Equation (4.18).

To better understand the nature and impact of the effective reflection coefficient of the substrate, let's consider two special cases where one layer (layer 3) lies between the resist and the substrate. For the first case, let this layer 3 be thick and absorbing enough so that no reflected light from the substrate makes it back to the top of layer 3. In other words, let $\tau_{D3}^2 = 0$. From the equations above, X_3 will equal 1 and $\rho'_{23} = \rho_{23}$. In other words, a sufficiently thick absorbing layer acts as a substrate. For the second case, assume layer 3 is optically matched to the photoresist so that $n_2 = n_3$. For this case, $\rho'_{23} = \rho_{34}\tau_{D3}^2$. Ignoring any small amount of absorption that may occur in layer 3 (the imaginary part of n_2 is usually very small), τ_{D3}^2 will have a magnitude of about 1 and will thus only impact the phase of ρ'_{23} . In other words, the magnitude of the effective substrate reflection coefficient will be determined by the reflection coefficient of layer 3 with the substrate and the phase of the effective reflection coefficient will be determined by the thickness of layer 3. (The resist on oxide on silicon case shown in Figure 4.3 is very similar to this idealized case.) This impact of the phase of the reflection coefficient on lithography is explored below in section 4.6.

If the thin film in question is not the top film (layer 2), the intensity can be calculated in layer j from

$$E_j(z) = E_{\text{Leff}} \tau_{j-1,j}^* \frac{(e^{ik_j z_j} + \rho'_{j,j+1} \tau_{Dj}^2 e^{-ik_j z_j})}{1 + \rho_{j-1,j}^* \rho'_{j,j+1} \tau_{Dj}^2} \quad (4.26)$$

where $\tau_{j-1,j}^* = 1 + \rho_{j-1,j}^*$. The effective reflection coefficient ρ^* is analogous to the coefficient ρ' , looking in the opposite direction. E_{Leff} is the effective intensity incident on layer

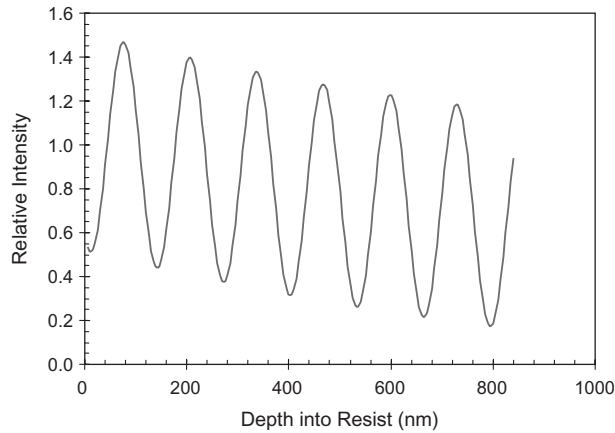


Figure 4.3 Standing wave intensity within a photoresist film at the start of exposure (850 nm of resist on 100-nm SiO_2 on silicon, $\lambda = 436 \text{ nm}$). Note the impact of the oxide film on the phase of the effective substrate reflectivity, which affects the intensity at the bottom of the resist

j. Both E_{leff} and ρ^* are defined in detail in Reference 3. Equation (4.26) is needed when using a top antireflection coating, as described in section 4.4.

If the resist film is not homogeneous, the equations above are, in general, not valid. Let us, however, examine one special case in which the inhomogeneity takes the form of small variations in the imaginary part of the index of refraction of the film in the z -direction, leaving the real part constant. Photoresist bleaching (described in detail in Chapter 5) will cause such a variation of the absorption coefficient with depth into the resist. In this case, the absorbance Abs is no longer simply αz , but becomes

$$Abs(z) = \int_0^z \alpha(z') dz' \quad (4.27)$$

Equations (4.18), (4.25) and (4.26) are still approximately valid if this anisotropic expression for absorbance is used. Thus, $I(z)$ can be found if the absorption coefficient is known as a function of z .

Figure 4.3 shows a typical result of the standing wave intensity within a photoresist film coated on an oxide on silicon film stack.

4.1.4 Oblique Incidence and the Vector Nature of Light

For simplicity, first consider a plane wave traveling through a uniform material and striking a partially reflective substrate at $z = 0$ (see Figure 4.4). The plane wave, described by its propagation vector $\mathbf{k}$ (defined as the propagation constant multiplied by a unit vector pointing in the direction of propagation), makes an angle θ with respect to the z -axis (which is also normal to the reflecting surface). Taking the x - z plane as the plane of incidence, any point in space (x, z) is described by its position vector $\mathbf{r}$. The electric field of the plane wave (before it reflects off the substrate) is then given by

$$E(x, z) = Ae^{i\mathbf{k} \cdot \mathbf{r}} = Ae^{ik(x \sin \theta + z \cos \theta)} \quad (4.28)$$

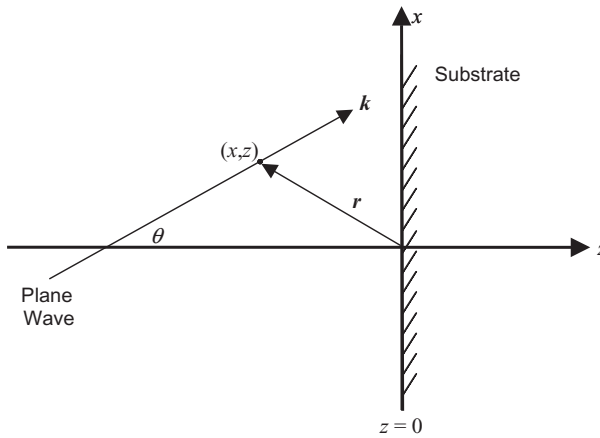


Figure 4.4 Geometry used for describing plane waves and standing waves for oblique incidence

where A is the amplitude of the plane wave and the (complex) propagation constant k is given by $2\pi n/\lambda$ where n is the complex refractive index of the medium. This plane wave will reflect off the substrate with an angle- and polarization-dependent reflectivity $\rho(\theta)$ as given by the Fresnel formulae.

$$\begin{aligned}
 \rho_{ij\perp}(\theta_j) &= \frac{n_i \cos(\theta_i) - n_j \cos(\theta_j)}{n_i \cos(\theta_i) + n_j \cos(\theta_j)} \\
 \tau_{ij\perp}(\theta_j) &= \frac{2n_i \cos(\theta_i)}{n_i \cos(\theta_i) + n_j \cos(\theta_j)} \\
 \rho_{ij\parallel}(\theta_j) &= \frac{n_i \cos(\theta_j) - n_j \cos(\theta_i)}{n_i \cos(\theta_j) + n_j \cos(\theta_i)} \\
 \tau_{ij\parallel}(\theta_j) &= \frac{2n_i \cos(\theta_i)}{n_i \cos(\theta_j) + n_j \cos(\theta_i)}
 \end{aligned} \tag{4.29}$$

and the two angles θ_i and θ_j are related by Snell's law,

$$n_i \sin(\theta_i) = n_j \sin(\theta_j) \tag{4.30}$$

Here, $\parallel$ represents an electric field vector that lies in a plane defined by the direction of the incident light and a normal to the resist surface (i.e. in the plane of the paper in Figure 4.4). Other names for $\parallel$ polarization include *p-polarization* and TM (transverse magnetic) polarization. The polarization denoted by $\perp$ represents an electric field vector that lies in a plane perpendicular to that defined by the direction of the incident light and a normal to the resist surface (i.e. perpendicular to the plane of the paper in Figure 4.4). Other names for $\perp$ polarization include *s-polarization* and TE (transverse electric) polarization. Note that for light normally incident on the resist surface, both *s*- and *p*-polarization result in electric fields that lie along the resist surface and the four Fresnel formulae revert to the two standard definitions of reflection and transmission coefficients used earlier in Equations (4.7) and (4.8). Figure 4.5 shows how the intensity reflectivity (the square of

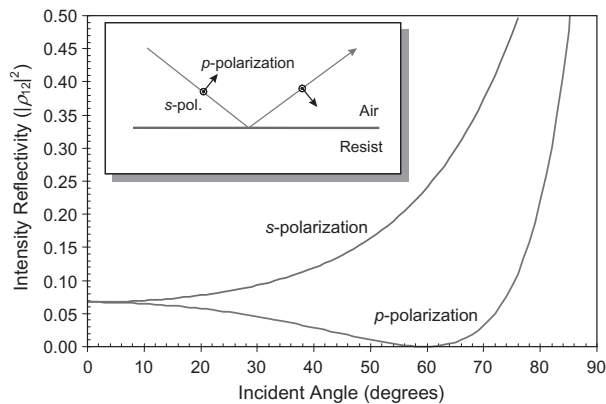


Figure 4.5 Reflectivity (square of the reflection coefficient) as a function of the angle of incidence showing the difference between *s*- and *p*-polarization ($n_1 = 1.0$, $n_2 = 1.7$). Both air and resist layers are assumed to be infinitely thick

the magnitude of the reflection coefficient) varies with incident angle for both *s*- and *p*-polarized illumination for a particular case.

The total electric field to the left of the substrate in Figure 4.4 is then the sum of the incident and reflected waves. This simple summation brings up the first interesting complication – the effect of polarization, the direction that the electric field is pointing. If the original electric field in Equation (4.28) is *s*-polarized (also called TE polarized) so that the electric field vector points directly out of the plane of Figure 4.4, then both the incident and reflected electric fields will point in the same direction and the vector sum will equal the scalar (algebraic) sum of the incident and reflected fields:

$$\begin{aligned} \text{s-polarized: } E(x, z) &= Ae^{ik_x \sin \theta} (e^{ik_z \cos \theta} + \rho(\theta) e^{-ik_z \cos \theta}) \\ I(x, z) &= |E(x, z)|^2 = |A|^2 (1 + |\rho(\theta)|^2 + 2|\rho(\theta)| \cos(2kz \cos \theta + \phi_\rho)) \end{aligned} \quad (4.31)$$

where ϕ_ρ is the phase angle of the complex reflectivity ρ and, for simplicity, the propagation medium is assumed to be nonabsorbing. The amplitude of the standing waves, given by the factor multiplying the cosine in the intensity expression above, is controlled by the magnitude of the electric field reflectivity. Note that this equation reverts to the simpler perfect mirror Equation (4.5) when $\rho = -1$.

Thus, for *s*-polarized light, the standing wave Equation (4.18) can be easily modified for the case of non-normally incident plane waves. Suppose a plane wave is incident on the resist film at some angle θ_1 . The angle of the plane wave inside the resist will be θ_2 as determined from Snell's law. An analysis of the propagation of this plane wave within the resist will give an expression similar to Equation (4.18) but with the position z replaced with $z \cos \theta_2$. For this case, the electric field will point in the *y*-direction.

$$\frac{E_y(z, \theta_2)}{E_i} = \frac{\tau_{12}(\theta_2)(e^{i2\pi n_2 z \cos \theta_2 / \lambda} + \rho_{23}(\theta_2)\tau_D^2(\theta_2)e^{-i2\pi n_2 z \cos \theta_2 / \lambda})}{1 + \rho_{12}(\theta_2)\rho_{23}(\theta_2)\tau_D^2(\theta_2)} \quad (4.32)$$

The transmission and reflection coefficients are now functions of the angle of incidence (as well as the polarization of the incident light) and are given by the Fresnel formulae above. The internal transmittance becomes

$$\tau_D = e^{i2\pi n_2 D \cos \theta_2 / \lambda} \quad (4.33)$$

The resulting relative intensity becomes

$$\begin{aligned} I(z) &= T_{\text{eff}}(\theta_2)(e^{-\alpha z / \cos \theta_2} + |\rho_{23}(\theta_2)|^2 e^{-\alpha(2D - z) / \cos \theta_2}) \\ &\quad + T_{\text{eff}}(\theta_2)2|\rho_{23}(\theta_2)|e^{-\alpha D / \cos \theta_2} \cos(4\pi n_2 \cos \theta_2 (D - z) / \lambda + \phi_{23}) \end{aligned} \quad (4.34)$$

(For an understanding of why the absorption terms vary as distance/cos θ_2 , see section 4.8.2)

For *p*-polarized incident light, reduced overlap in electric fields causes the standing wave amplitude to be reduced by the factor $\cos(2\theta_2)$, as will be shown below.⁴ The resulting electric field will have both *x*- and *z*-components (*x* being the dimension in the plane of incidence), whereas for the *s*-polarized case the electric field only points in the *y*-direction (out of the plane of incidence).

$$\frac{E_x}{E_i} = \frac{\cos(\theta_2)\tau_{12}(\theta_2)(e^{i2\pi n_2 z \cos \theta_2 / \lambda} + \rho_{23}(\theta_2)\tau_D^2(\theta_2)e^{-i2\pi n_2 z \cos \theta_2 / \lambda})}{1 + \rho_{12}(\theta_2)\rho_{23}(\theta_2)\tau_D^2(\theta_2)} \quad (4.35)$$

$$\frac{E_z}{E_1} = \frac{\sin(\theta_2)\tau_{12}(\theta_2)(e^{i2\pi n_2 z \cos \theta_2 / \lambda} - \rho_{23}(\theta_2)\tau_D^2(\theta_2)e^{-i2\pi n_2 z \cos \theta_2 / \lambda})}{1 + \rho_{12}(\theta_2)\rho_{23}(\theta_2)\tau_D^2(\theta_2)} \quad (4.36)$$

The total standing wave intensity, relative to the incident intensity, will be

$$\begin{aligned} I(z) &= \frac{n_2(|E_x|^2 + |E_z|^2)}{n_1|E_1|^2} \\ &= T_{\text{eff}}(\theta_2)(e^{-\alpha z / \cos \theta_2} + |\rho_{23}(\theta_2)|^2 e^{-\alpha(2D-z) / \cos \theta_2}) \\ &\quad + T_{\text{eff}}(\theta_2)2|\rho_{23}(\theta_2)|e^{-\alpha D / \cos \theta_2} \cos(2\theta_2) \cos(4\pi n_2 \cos \theta_2 (D-z) / \lambda + \phi_{23}) \end{aligned} \quad (4.37)$$

If $\theta_2 = 45^\circ$, the incident and reflected waves have no overlap of their electric fields and the resulting lack of interference means there will be no standing waves (only the bulk-effect term survives).

It is interesting to look at the impact of direction that the light is traveling on the definitions of the reflection and transmission coefficients. Completely reversing the direction of the light, if light approaches the interface through material j at an angle θ_j , the resulting reflection and transmission coefficients become

$$\begin{aligned} \rho_{ji} &= -\rho_{ij} \\ \tau_{ji} &= \frac{n_j \cos(\theta_j)}{n_i \cos(\theta_i)} \tau_{ij} \end{aligned} \quad (4.38)$$

where these relationships hold for either polarization.

Figure 4.6 shows an example of the impact of incident angle on the resulting standing wave pattern in the resist. The change in incident angle causes a change in the period of the standing waves and, because of the angular dependence of T_{eff} , a difference in the amount of light transmitted into the photoresist film.

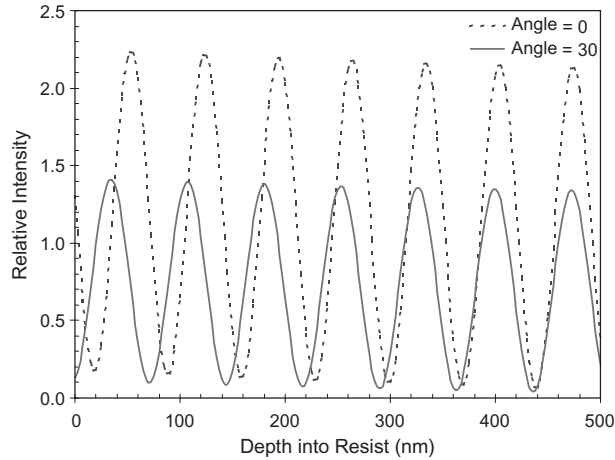


Figure 4.6 Standing wave intensity within a photoresist film (500nm of resist on silicon, $\lambda = 248\text{nm}$) as a function of incident angle (s-polarization assumed)

A second approximation can be made to simplify the evaluation of Equations (4.32), (4.35) or (4.36). For a complex resist refractive index, Snell's law produces the nonintuitive but mathematically correct result of a complex angle of refraction in the resist. For the case where the resist is only weakly absorbing (quite a normal situation), the imaginary part of the angle of refraction into the resist can be ignored with only a small loss of accuracy. The impact of making such an approximation is explored more fully below in section 4.8.

4.1.5 Broadband Illumination

The analysis presented above applies to monochromatic illumination. What if broadband (polychromatic) illumination were used? In general, the use of broadband illumination in imaging applications can be thought of as the incoherent superposition of individual monochromatic results. Thus, the standing wave intensity for a single wavelength is integrated over the wavelengths of the source, weighted by the source illumination spectrum. Figure 4.7 shows a typical mercury arc lamp output spectrum (before filtering in the illumination system of the projection tool). Table 4.1 describes the common wavelengths used in lithographic tools. For 248- and 193-nm lithography tools, where a line-narrowed excimer laser is used to generate the light, bandwidths are less than a picometer, so that these sources can be considered monochromatic from a standing wave perspective.

Although wavelength appears explicitly in the expressions for the standing wave electric field in the resist, an implicit dependence occurs in the indices of refraction of the various resist and substrate materials. The variation of the index of refraction of a material with wavelength, called dispersion, is extremely material dependent. Over a limited range of wavelengths, many photoresist materials' dispersion curves can be adequately described

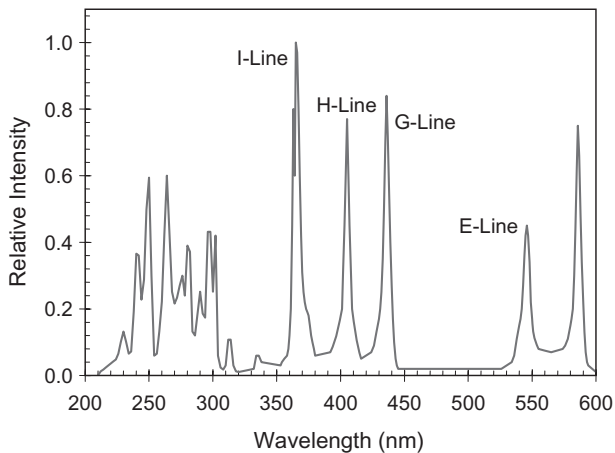


Figure 4.7 Spectral output of a typical high-pressure mercury arc lamp. The illumination spectrum of an i-line or g-line lithographic exposure tool is usually a filtered portion of this lamp spectrum

Table 4.1 Wavelengths for various lithographic light sources (high pressure mercury arc lamps and free-running, unnarowed excimer lasers)

Source	Center Wavelength (nm)	Typical Unnarowed Bandwidth (nm)
Mercury Arc Lamp g-line	435.8	5
Mercury Arc Lamp h-line	404.7	5
Mercury Arc Lamp i-line	365.0	6
KrF excimer laser	248.35	0.30
ArF excimer laser	193.3	0.45
F ₂ excimer laser	157.63	0.002

Table 4.2 Some values for Cauchy coefficients of photoresists (from <http://www.microe.rit.edu/research/lithography/>)

Photoresist	C ₁	C ₂ (nm ²)	C ₃ (nm ⁴)	Wavelength Range (nm)
SPR-500 i-line (unexposed)	1.6133	5481.6	1.4077×10^9	300–800
SPR-500 i-line (fully exposed)	1.5954	9291.9	4.2559×10^8	300–800
193-nm resist	1.5246	3484	1.49×10^8	190–800

by an empirical expression called the Cauchy equation (named for the 19th century French mathematician Augustin-Louis Cauchy):

$$n(\lambda) = C_1 + \frac{C_2}{\lambda^2} + \frac{C_3}{\lambda^4} \quad (4.39)$$

where C_1 , C_2 and C_3 are the empirically derived Cauchy coefficients (higher-order terms are possible, but rarely used for photoresists). It is important to note that any Cauchy fit to refractive index data will be valid only over the wavelength range of the data and should not be extrapolated. Further, this equation applies well only to dielectrics with no or small amounts of absorption over the wavelength range of interest. Some typical resist Cauchy coefficients are given in Table 4.2.

Figure 4.8 shows the impact of using a range of wavelengths on the resulting standing wave pattern in the resist. In this example, a single 365-nm-wavelength exposure is compared to a very broadband exposure over a range of wavelengths from 350 to 450 nm using a mercury arc lamp exposure tool. The total broadband standing wave result is the sum of many monochromatic standing wave curves, each with different periods due to the different wavelengths. As a result, these sinusoids tend to average out making the final intensity within the film more uniform with depth. Note also that the standing waves are at their largest at the substrate, where an insufficient number of standing wave periods are available for averaging.

An alternate viewpoint is to consider the *coherence length* of the light source. While commonly used to describe lasers, coherence length can in fact be used to describe any light source and is defined as the longest distance light can travel and still maintain a

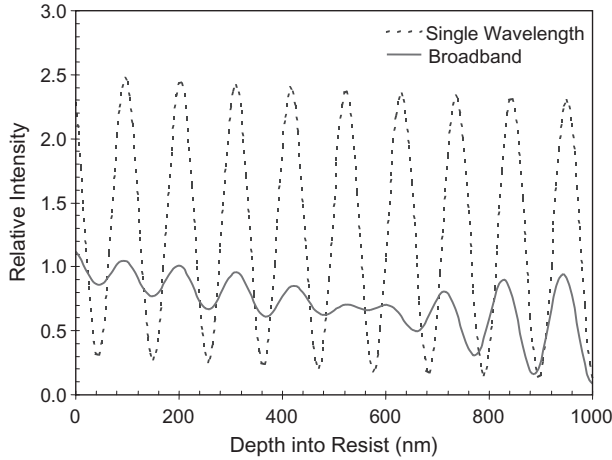


Figure 4.8 Standing wave intensity within a photoresist film (1000 nm of resist on silicon), for monochromatic ($\lambda = 365$ nm) and broadband illumination (350–450 nm range of the mercury spectrum)

fixed phase relationship all along that length (that is, still maintain the ability to interfere with itself). Suppose light of wavelength λ_1 travels a distance $L/2$, strikes a mirror at normal incidence, and travels back a distance $L/2$. Further suppose that the total distance traveled L is an integer number of wavelengths $N\lambda_1$. Consider now a second wavelength λ_2 such that this same distance is equal to $(N + 1/2)\lambda_2$. For this case, when the first wavelength interferes constructively, the second wavelength will be interfering destructively. If both wavelengths are present, no interference pattern will be observed at the distance $L/2$ away from the mirror and L is called the coherence length.

Combining the two equations

$$L = N\lambda_1 = (N + 1/2)\lambda_2 = \left(\frac{L}{\lambda_1} + 1/2\right)\lambda_2 \quad (4.40)$$

Solving for L ,

$$L = \frac{\lambda_1 \lambda_2}{2(\lambda_1 - \lambda_2)} \quad (4.41)$$

Letting $\lambda_1 - \lambda_2 = \Delta\lambda$, the bandwidth, and assuming this bandwidth is small compared to the wavelength so that $\lambda_1 \lambda_2 = \lambda^2$, where the geometric mean λ is about equal to the arithmetic mean, the equation becomes

$$L = \frac{\lambda^2}{2\Delta\lambda} \quad (4.42)$$

As an example, if both g-line and i-line light exposes a photoresist, the range of wavelengths is about 80 nm and the mean wavelength is about 400 nm. The coherence length will be 1000 nm, which means that there will be no standing waves at about 500 nm up

from the substrate. This matches the result seen in Figure 4.8 quite well. For laser lithography at 248 or 193 nm, the source bandwidth is extremely small. For a 248-nm source with a 1-pm bandwidth, the coherence length will be on the order of 3 cm.

4.2 Swing Curves

Generically, a swing curve refers to the sinusoidal variation of some lithographic parameter with resist thickness.⁵ There are several parameters that vary in this way, but the most important is the critical dimension (CD) of the photoresist feature being printed. Figure 4.9 shows a typical CD swing curve for i-line exposure of a 0.5- μm line on silicon. The change in linewidth is quite large (more than the typical 10% tolerance) for relatively small changes in resist thickness. Another swing curve is the E_0 swing curve, showing the same sinusoidal swing in the photoresist dose-to-clear (Figure 4.10). For a resist thickness that requires a higher dose-to-clear, the photoresist will, as a consequence, require a higher dose to achieve the desired line size. But if the exposure dose is fixed (as it was for the CD swing curve), the result will be an underexposed line that prints too large. Thus, it follows that the E_0 and CD swing curves result from the same effect, coupled by the exposure latitude of the feature. The final swing curve measures the reflectivity of the resist-coated wafer as a function of resist thickness (Figure 4.11). Although reflectivity is further removed from lithographic metrics such as E_0 or CD, it is the reflectivity swing curve that provides the most insight as to the cause of the phenomenon.

4.2.1 Reflectivity Swing Curve

The reflectivity swing curve shows that variations in resist thickness result in a sinusoidal variation in the reflectivity of the resist-coated wafer. Since the definition of reflectivity is the total reflected light intensity divided by the total incident intensity, an increase in

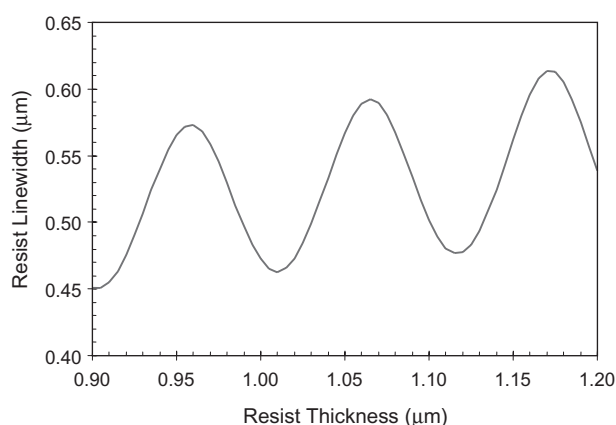


Figure 4.9 CD swing curve showing a sinusoidal variation in the resist linewidth with resist thickness (i-line exposure of resist on silicon)

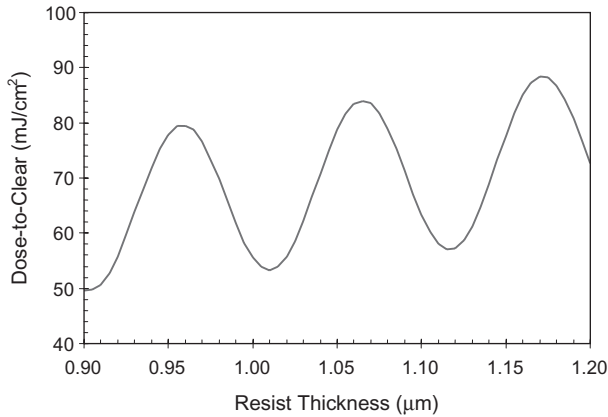


Figure 4.10 E_0 swing curve showing a sinusoidal variation in the resist dose-to-clear with resist thickness (i-line exposure of resist on silicon)

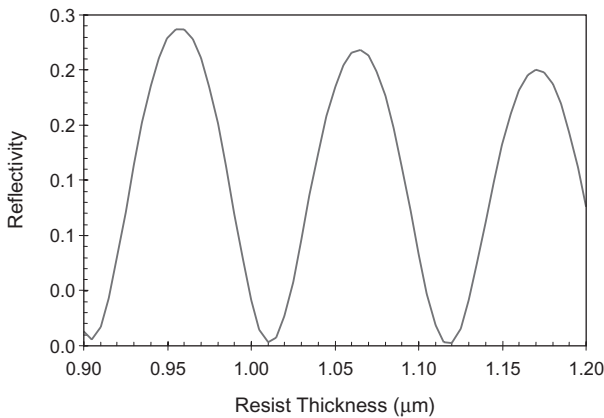


Figure 4.11 Reflectivity swing curve showing a sinusoidal variation in the resist-coated wafer reflectivity with resist thickness (i-line exposure of resist on silicon)

reflectivity results in more light that does not make it into the resist. Less light being coupled into the resist means that a higher dose is required to affect a certain chemical change in the resist, resulting in a larger E_0 . Thus, the E_0 and CD swing curves can both be explained by the reflectivity swing curve. (The interested reader can convince him/herself that the phases of the sinusoids of Figure 4.9 through Figure 4.11 make sense with respect to each other.)

What causes the reflectivity swing curve of Figure 4.11? Of course, the answer lies in the thin film interference effects that were discussed in the previous section on standing waves. Using the same simple geometry shown in Figure 4.1, a thin photoresist (layer 2) rests on a thick substrate (layer 3) in air (layer 1). If we illuminate this film stack with a

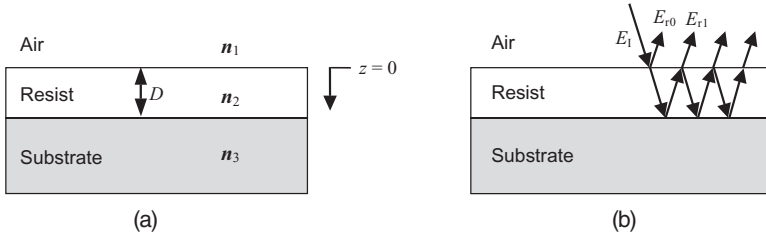


Figure 4.12 Film stack showing (a) geometry for swing curve derivation, and (b) incident, transmitted and reflected waves [oblique angles are shown for diagrammatical purposes only]

monochromatic plane wave normally incident on the resist, the analysis given before was used to determine the standing wave intensity within the resist. However, our goal here is to determine the total light reflected by the film stack. As shown in Figure 4.12, the total reflected light is made up of the incident beam reflecting off the air–resist interface and beams that have bounced off of the substrate and then were transmitted by the air–resist interface.

Let's begin by writing an expression for the electric field of the ray that is directly reflected by the air–resist interface. Recalling the definitions used in section 4.1.2,

$$E_{r0} = \rho_{12} E_i \quad (4.43)$$

The next 'reflected' beam is transmitted into the resist, reflected off the substrate, and transmitted into the air. The result, denoted as E_{r1} , is given by

$$E_{r1} = E_i \tau_{12} \tau_{21} \rho_{23} \tau_D^2 \quad (4.44)$$

The next reflected beam makes two bounces inside the resist before being transmitted out, resulting in an additional $\rho_{21} \rho_{23} \tau_D^2$ term.

The total reflection coefficient can be computed by totaling up all the reflected electric fields and then dividing by the incident field.

$$\rho_{\text{total}} = \frac{\sum E_{ri}}{E_i} = \rho_{12} + \frac{\tau_{12} \tau_{21} \rho_{23} \tau_D^2}{1 + \rho_{12} \rho_{23} \tau_D^2} = \frac{\rho_{12} + \rho_{23} \tau_D^2}{1 + \rho_{12} \rho_{23} \tau_D^2} \quad (4.45)$$

Note that since all of the reflected light will always be traveling in the same direction, Equation (4.45) applies equally well to *s*- or *p*-polarization. The intensity reflectivity of the film stack is the square of the magnitude of the electric field reflection coefficient. At first glance, the sinusoidal dependence of reflectivity with resist thickness is not obvious from Equation (4.45). The dependence is contained in the internal transmittance. Carrying out the calculation of reflectivity for the case of normal incidence,

$$R = \frac{|\rho_{12}|^2 + |\rho_{23}|^2 e^{-\alpha 2D} + 2|\rho_{12} \rho_{23}| e^{-\alpha D} \cos(4\pi n_2 D / \lambda - \phi_{12} + \phi_{23})}{1 + |\rho_{12} \rho_{23}|^2 e^{-\alpha 2D} + 2|\rho_{12} \rho_{23}| e^{-\alpha D} \cos(4\pi n_2 D / \lambda + \phi_{12} + \phi_{23})} \quad (4.46)$$

where ϕ_{12} and ϕ_{23} are the phase angles of the reflection coefficients ρ_{12} and ρ_{23} , respectively.

The discussion so far has been mostly mathematical. Equation (4.45) gives a rigorous result that, when expressed as Equation (4.46), leads to an understanding of the reflectivity swing curve. Physically, the reflectivity swing curve is the result of interference among the reflected rays. As pictured in Figure 4.12, the total reflected field is the sum of the various rays. How the initially reflected ray E_{r0} adds to the first transmitted and reflected ray E_{r1} depends on the phase of E_{r1} , which in turn depends on the resist thickness. At some thickness, E_{r1} will be in phase with E_{r0} , resulting in a maximum reflectivity. At another thickness E_{r1} will be out of phase with E_{r0} , resulting in a minimum reflectivity.

Equation (4.46) can also lead to a better understanding of swing curves. The period of all of the swing curves can be easily obtained from Equation (4.46) and is the same as the period of the standing waves in the photoresist. Likewise, the effects of increasing or reducing the reflectivities can be seen. Twice the multiplier of the cosine in Equation (4.46) is called the *swing amplitude*,⁶ and is equal to $4|\rho_{12}\rho_{23}|e^{-\alpha d}$. If the substrate is non-reflective ($\rho_{23} = 0$), the film stack reflectivity becomes constant (the swing amplitude goes to zero). Thus, a bottom antireflection coating can reduce or eliminate the swing curve. Less obviously, if $\rho_{12} = 0$ the reflectivity will also become constant, eliminating the swing curve. This can be achieved by using a top antireflection coating. Physically, if the swing curve results from interference between E_{r0} and E_{r1} , eliminating either E_{r0} or E_{r1} will eliminate the interference and the swing. Finally, absorption in the resist will reduce the swing amplitude in Equation (4.46) by reducing the amplitude of E_{r1} and thus the amount of interference that can take place.

While not as obvious from the above discussion, the use of broadband illumination can reduce or eliminate swing curves as well. Recalling the discussion in section 4.1.5, swing curves will be reduced whenever the resist thickness approaches the coherence length of the light source.

Some refractive index values for common materials encountered in optical lithography are given in Table 4.3. For the case of a typical resist on silicon at 248 nm, $\alpha = 0.5 \mu\text{m}^{-1}$, $|\rho_{12}| = 0.275$ and $|\rho_{23}| = 0.73$. For a nominal 300-nm resist thickness, that gives

$$R = \frac{0.47 + 0.345 \cos(2\pi D / (70.5 \text{ nm}) - \phi_{12} + \phi_{23})}{1.03 + 0.345 \cos(2\pi D / (70.5 \text{ nm}) + \phi_{12} + \phi_{23})} \quad (4.47)$$

Table 4.3 Some refractive index values of common materials at common lithographic wavelengths⁷

Material	n at 436 nm	n at 365 nm	n at 248 nm	n at 193 nm
Photoresist	$1.65 + i0.022$	$1.69 + i0.027$	$1.76 + i0.010$	$1.71 + i0.018$
Silicon	$4.84 + i0.178$	$6.50 + i2.61$	$1.57 + i3.57$	$0.883 + i2.78$
Amorphous Silicon	$4.45 + i1.73$	$3.90 + i2.66$	$1.69 + i2.76$	$1.13 + i2.10$
Silicon Dioxide	$1.470 + i0.0$	$1.474 + i0.0$	$1.51 + i0.0$	$1.56 + i0.0$
Silicon Nitride	$2.06 + i0.0$	$2.09 + i0.0$	$2.28 + i0.0$	$2.66 + i0.240$
Aluminum	$0.580 + i5.30$	$0.408 + i4.43$	$0.190 + i2.94$	$0.113 + i2.20$
Copper	$1.17 + i2.33$	$1.27 + i1.95$	$1.47 + i1.78$	$0.970 + i1.40$
Chrome	$1.79 + i4.05$	$1.39 + i3.24$	$0.85 + i2.01$	$0.84 + i1.65$
Gallium Arsenide	$5.07 + i1.25$	$3.60 + i2.08$	$2.27 + i4.08$	$1.36 + i2.02$
Indium Phosphide	$4.18 + i0.856$	$3.19 + i1.95$	$2.13 + i3.50$	$1.49 + i2.01$
Germanium	$4.03 + i2.16$	$4.07 + i2.58$	$1.39 + i3.20$	$1.13 + i2.09$

where $\phi_{12} = -179.5^\circ$ and $\phi_{23} = -134^\circ$. The maximum reflectivity is about 0.59, and the minimum is about 0.18. For the case of a typical resist on silicon at 365 nm, $\alpha = 0.95 \mu\text{m}^{-1}$, $|\rho_{12}| = 0.257$ and $|\rho_{23}| = 0.634$ (before bleaching). For a nominal 600-nm resist thickness, that gives

$$R = \frac{0.195 + 0.184 \cos(2\pi D/(108 \text{ nm}) - \phi_{12} + \phi_{23})}{1.008 + 0.184 \cos(2\pi D/(108 \text{ nm}) + \phi_{12} + \phi_{23})} \quad (4.48)$$

The maximum reflectivity for this case is about 0.318, and the minimum is about 0.013.

4.2.2 Dose-to-Clear and CD Swing Curves

An approximate behavior of the E_0 swing curve can be obtained by assuming an ideal, threshold resist. For such a case, the resist will just clear away when the dose at the bottom of the resist (for an open-frame exposure) reaches some critical threshold. Using Equation (4.19) and assuming the standing waves in the resist are averaged out by some post-exposure bake mechanism (see Chapter 5), the average or bulk intensity will be

$$I_{\text{avg}}(z) = T_{\text{eff}}(e^{-\alpha z} + |\rho_{23}|^2 e^{-\alpha(2D-z)}) \quad (4.49)$$

where this equation was derived by averaging the standing wave intensity Equation (4.19) over one period. It is the effective transmittance T_{eff} that contains the swing curve effect of energy coupled into the resist as a function of resist thickness. Evaluating this expression at the bottom of the resist,

$$I_{\text{avg}}(z = D) = T_{\text{eff}}(1 + |\rho_{23}|^2)e^{-\alpha D} \quad (4.50)$$

The incident dose will equal the dose-to-clear, E_0 , when the energy at the bottom of the resist (equal to $E_0 I_{\text{avg}}$) reaches some critical dose, E_{crit} . Thus,

$$\begin{aligned} E_0 &= \frac{E_{\text{crit}}}{T_{\text{eff}}(1 + |\rho_{23}|^2)e^{-\alpha D}} \\ &= E_{\text{crit}} \frac{1 + |\rho_{12}\rho_{23}|^2 e^{-\alpha 2D} + 2|\rho_{12}\rho_{23}|e^{-\alpha D} \cos(4\pi n_2 D/\lambda + \phi_{12} + \phi_{23})}{T_{12}(1 + |\rho_{23}|^2)e^{-\alpha D}} \end{aligned} \quad (4.51)$$

Likewise, for a threshold resist, the dose to achieve a certain CD will follow this same equation though with a different critical dose. For a fixed dose, the CD obtained will depend on the exposure latitude of the feature. A simple expression that relates CD to exposure dose will be derived in Chapter 8, and takes the form

$$\frac{CD - CD_1}{CD_1} = \frac{d \ln CD}{d \ln E} \bigg|_{E=E_1} \left(1 - \frac{E_1}{E} \right) \quad (4.52)$$

where E_1 is a reference dose (such as the nominal dose) that produces the $CD = CD_1$, and the exposure latitude term ($d \ln CD / d \ln E$) can be approximated as a constant.

In the context of swing curves, the best choice for E_1 will be the dose that produces the nominal CD when the resist thickness is at a minimum or maximum. For the case of operating at a swing curve minimum (resist thickness = $D_{\min}$),

$$E_1 = E_{\text{crit}} \frac{1 + |\rho_{12}\rho_{23}|^2 e^{-\alpha 2D_{\min}} - 2|\rho_{12}\rho_{23}| e^{-\alpha D_{\min}}}{T_{12}(1 + |\rho_{23}|^2) e^{-\alpha D_{\min}}} \quad (4.53)$$

Substituting Equation (4.51) for E and Equation (4.53) for E_1 into Equation (4.52) gives a CD swing curve expression.

4.2.3 Swing Curves for Partially Coherent Illumination

Just as for the standing wave expressions, the above swing curve equations will change as a function of the incident angle of illumination, so that actual swing curves are strongly dependent on the NA and illumination of the stepper. In fact, the range of angles coming from a partially coherent imaging source will produce a range of periods of standing waves and swing curves in the resist, reducing the amplitude in the same way that a range of wavelengths does.

Suppose that the incident light strikes the resist at an angle θ_1 . The angle inside the resist will be θ_2 , given by Snell's law. Then the swing curve expressions given above will be correct if the resist thickness D is replaced by $D \cos \theta_2$ and the reflection and transmission coefficients are calculated for the appropriate angle and polarization according to the Fresnel formulae [Equation (4.29)]. The swing curve period becomes

$$\text{Period} = \frac{\lambda}{2n_2 \cos \theta_2} \quad (4.54)$$

Consider a simple case. Light normally incident on a thin resist film coated on a reflective substrate is found to have a maximum of its swing curve at a certain thickness. This means that the path length traveled by the light through the resist is an integer multiple of the wavelength. If now the angle of incidence is increased, the path length through the resist will also increase. If the angle is large enough so that the path length increases by half a wavelength, this same resist thickness will correspond to a minimum of the swing curve. The angle inside the resist which causes this swing curve phase reversal is given by

$$\theta_2 = \cos^{-1} \left(\frac{m}{m + 0.5} \right) \quad (4.55)$$

where m is the multiple of wavelengths that the swing curve maximum represents. For $m = 9$ (typical for a 1000-nm film used for i-line lithography), $\theta_2 \approx 18.7^\circ$ inside the resist, corresponding to about a 32° angle incident on a typical photoresist and $\sin \theta_1 = 0.53$. For $m = 5$, (typical for a 250-nm resist film used for 193-nm lithography), $\theta_2 \approx 24.6^\circ$ inside the resist, corresponding to about a 45° angle incident on a typical photoresist and $\sin \theta_1 = 0.71$. Both of these conditions are commonly met.

What does this mean for real swing curves in real lithographic situations? When measuring a dose-to-clear (E_0) swing curve, the light striking the resist is made up of

zero-order light (that is, light which is not diffracted). If the illumination were coherent, the light would be normally incident on the resist. For conventional partially coherent illumination, the light is incident on the resist over a range of angles given by $\sin \theta_1 = \pm \sigma NA$ where NA is the numerical aperture of the objective lens and σ is the partial coherence factor. Each angle produces its own ‘swing curve’ and the total result is the superposition of all the individual responses to each angle. Since any change in σ or NA will change the range of angles striking the photoresist, the swing curve will change as well. Figure 4.13 shows how E_0 swing curves vary with σNA (affected by changing either the partial coherence or the numerical aperture).

Consider also a simple example of imaging small lines and spaces. For conventional illumination, the zero order will be centered around normal incidence at the resist surface with a range of angles determined by σNA . The ± 1 st diffraction orders will strike the resist at an angle of $\sin^{-1}(\lambda/p)$ where p is the pitch of the line/space pattern. For 350-nm features imaged with i-line, the center of the first-order angular range will be about 31.4° in air – very close to the angle given above for swing curve phase reversal for a 1000-nm-thick resist film. For 130-nm lines and spaces imaged with 193-nm light, the first-order light also produces a swing that is nearly perfectly out of phase with the normally incident light. Thus, if the resist thickness were adjusted to give a maximum of the E_0 swing curve (i.e. the zero order is at a maximum of the swing curve), the first orders would effectively be at a minimum of the swing curve. The zero-order light would be maximally reflected out of the resist while the first-order light would be maximally coupled into the resist. When these orders combine to form the image in resist, the result will be significantly different than the case of imaging on a nonreflecting substrate. On the other hand, if the resist thickness were at an E_0 swing curve minimum, the first orders would be at a swing curve maximum. The lithographic response of these features (for example, the size of the focus-exposure process window) could be quite different when operating at an E_0 swing curve minimum versus a maximum.

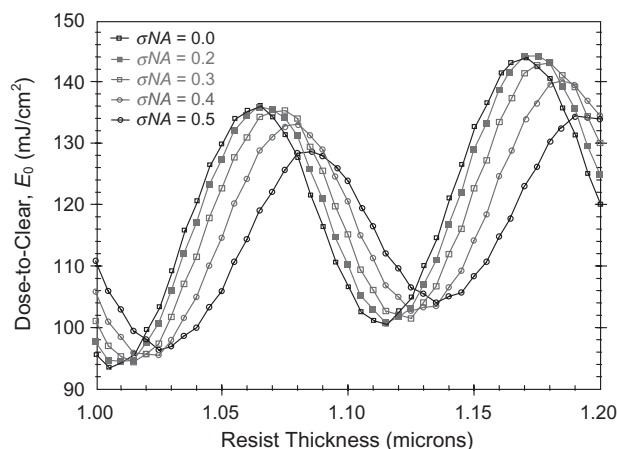


Figure 4.13 The phase and amplitude of a dose-to-clear swing curve are affected by the range of angles striking the resist, which is controlled by the product of the partial coherence and the numerical aperture (σNA) for conventional illumination

4.2.4 Swing Ratio

While detailed equations for swing curve effects are certainly possible, as the above discussions have shown, it is often convenient to define simple metrics to capture the essence of the effects. For swing curves, it is the amplitude of the swing that determines the degree of the problem caused by resist thickness variations. A simple metric for the magnitude of the swing is called the swing ratio, defined as the difference between adjacent min and max values of the swing curve divided by their average value. For example, for an E_0 swing curve, the swing ratio (SR) would be

$$SR = \frac{E_{0\max} - E_{0\min}}{(E_{0\min} + E_{0\max})/2} \quad (4.56)$$

Since any two adjacent min/max pairs can be used, it will make a difference whether the max is at a greater or lesser thickness compared to the min, as will be seen below.

Using Equation (4.51),

$$\begin{aligned} E_{0\min} &= E_{\text{crit}} \frac{e^{\alpha D_{\min}} + |\rho_{12}\rho_{23}|^2 e^{-\alpha D_{\min}} - 2|\rho_{12}\rho_{23}|}{T_{12}(1 + |\rho_{23}|^2)} \\ E_{0\max} &= E_{\text{crit}} \frac{e^{\alpha D_{\max}} + |\rho_{12}\rho_{23}|^2 e^{-\alpha D_{\max}} + 2|\rho_{12}\rho_{23}|}{T_{12}(1 + |\rho_{23}|^2)} \end{aligned} \quad (4.57)$$

which gives a swing ratio of

$$SR = \frac{(e^{\alpha D_{\max}} - e^{\alpha D_{\min}}) + |\rho_{12}\rho_{23}|^2 (e^{-\alpha D_{\max}} - e^{-\alpha D_{\min}}) + 4|\rho_{12}\rho_{23}|}{\frac{1}{2}[(e^{\alpha D_{\max}} + e^{\alpha D_{\min}}) + |\rho_{12}\rho_{23}|^2 (e^{-\alpha D_{\max}} + e^{-\alpha D_{\min}})]} \quad (4.58)$$

Let $D_{\max} = D + \Delta D/2$ and $D_{\min} = D - \Delta D/2$, where a positive value of ΔD means that the max is at a resist thickness greater than the min. (Note that ΔD will be the swing period, that is $\lambda/2n_2$.) Equation (4.58) can be rewritten as

$$SR = \frac{e^{\alpha D}(e^{\alpha \Delta D/2} - e^{-\alpha \Delta D/2}) + |\rho_{12}\rho_{23}|^2 e^{-\alpha D}(e^{-\alpha \Delta D/2} - e^{\alpha \Delta D/2}) + 4|\rho_{12}\rho_{23}|}{\frac{1}{2}[e^{\alpha D}(e^{\alpha \Delta D/2} + e^{-\alpha \Delta D/2}) + |\rho_{12}\rho_{23}|^2 e^{-\alpha D}(e^{-\alpha \Delta D/2} + e^{\alpha \Delta D/2})]} \quad (4.59)$$

If the amount of absorption is moderate, so that $\alpha \Delta D \ll 1$, this equation can be simplified to

$$SR \approx \frac{e^{\alpha D}(\alpha \Delta D) + |\rho_{12}\rho_{23}|^2 e^{-\alpha D}(-\alpha \Delta D) + 4|\rho_{12}\rho_{23}|}{e^{\alpha D} + |\rho_{12}\rho_{23}|^2 e^{-\alpha D}} \quad (4.60)$$

In general,

$$|\rho_{12}\rho_{23}|^2 e^{-2\alpha D} \ll 1 \quad (4.61)$$

(for example, for resist on silicon at 248 nm, the value of this term is about 0.01). Neglecting this term gives

$$SR \approx \alpha \Delta D + 4|\rho_{12}\rho_{23}|e^{-\alpha D} \quad (4.62)$$

Thus, the swing ratio is equal to the swing amplitude plus an absorption term that describes the tilt of the swing curve over one swing period.

From Equation (4.62) it is easy to see how the standard approaches to reducing swing curves are captured by the swing ratio metric: (1) reduce substrate reflectivity $|\rho_{23}|$ by using a bottom antireflection coating; (2) reduce air–resist reflectivity $|\rho_{12}|$ by using a top antireflection coating; and (3) reduce $e^{-\alpha D}$ by increasing absorption. However, the sign of ΔD will also have an impact. If the maximum of the swing curve is at a greater thickness than the minimum, ΔD will be positive and the upward tilt of the swing curve due to absorption will make the swing ratio worse. If, however, the maximum of the swing curve is chosen to be at a lower resist thickness than the minimum, ΔD will be negative and bulk absorption will lower the swing ratio. In fact, it is possible to make the swing ratio go to zero:

$$SR \approx 0 \quad \text{when} \quad \alpha = \frac{4|\rho_{12}\rho_{23}|e^{-\alpha D}}{-\Delta D} \quad (4.63)$$

Figure 4.14 shows an example of zero swing ratio for a specific case. The upward tilt of the swing curve due to absorption can make the swings look more like stair steps. Operating in the flat region (where $SR \approx 0$) can lead to greater CD control for these cases.

As mentioned in the previous section, light strikes the wafer over a range of angles, each angle setting up its own swing curve with its own period. If the range of angles is great enough, a very noticeable decrease in the total swing ratio can be observed (see Figure 4.13). For a simple case, we can calculate the impact of the range of angles on the swing ratio metric defined above. Let's consider the E_0 swing curve for conventional illumination. Further, let's assume that the effect of angle on the reflection and

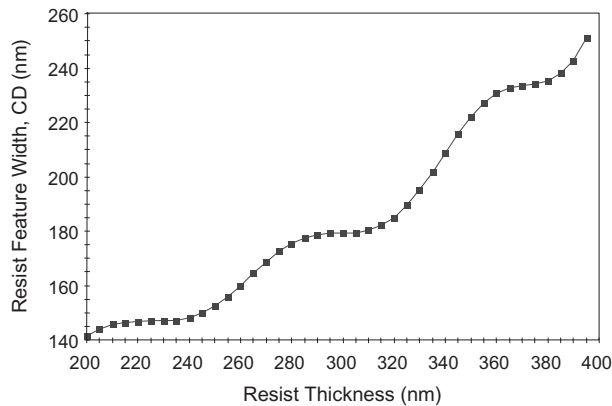


Figure 4.14 Proper balancing of absorption and reflectivities can make the minimum of a swing curve ($D = 310\text{ nm}$) achieve the same CD as the previous swing curve maximum ($D = 280\text{ nm}$)

transmission coefficients and on absorption is small compared to the effect on the period. Thus, we can put the E_0 swing curve formula [Equation (4.51)] into the basic approximate form

$$E_0 = a + b \cos(4\pi n_2 D \cos \theta_2 / \lambda + \phi) \quad (4.64)$$

where the swing ratio is $2b/a$.

Inside the resist, let $s = \sin \theta_2$ so that we must integrate over the range $\pm s_{\max}$ where

$$s_{\max} = \frac{\sigma NA}{n_2} \quad (4.65)$$

Integrating over the source,

$$E_0 = a + b \frac{2}{s_{\max}^2} \int_0^{s_{\max}} \cos(4\pi n_2 D \sqrt{1-s^2} / \lambda + \phi) s ds \quad (4.66)$$

This integral is not analytically solvable, but it can be approximated by letting

$$\cos \theta_2 = \sqrt{1-s^2} \approx 1 - \frac{s^2}{2} \quad (4.67)$$

A typical value for $s_{\max}$ is about 0.5, so that this approximation is not too bad. Using the approximation and carrying out the integral gives

$$E_0 = a + b \frac{\sin(c)}{c} \cos\left(4\pi n_2 D \left(1 - \frac{s_{\max}^2}{4}\right) / \lambda + \phi\right) \quad (4.68)$$

where

$$c = \frac{\pi D (\sigma NA)^2}{\lambda n_2}$$

By using the same small angle approximation as was used in Equation (4.67),

$$E_0 = a + b \frac{\sin(c)}{c} \cos\left(4\pi n_2 D \cos\left(\frac{\theta_{2\max}}{\sqrt{2}}\right) / \lambda + \phi\right) \quad (4.69)$$

At first glance, it is not clear what the period of the new standing wave is, since there is a product of a sine and a cosine. However, the sine term has a period that is 10–20 times larger than the cosine term, so it is easiest to interpret the cosine as creating the swing curve and the sinc function as an amplitude modification. Thus, the period of the partially coherent swing curve is about

$$\text{Period} = \frac{\lambda}{2n_2 \cos \theta_{\text{eff}}} \quad \text{where} \quad \theta_{\text{eff}} = \frac{\theta_{2\max}}{\sqrt{2}} \quad (4.70)$$

and the swing ratio is reduced by the factor $\sin(c)/c$. Consider a typical 193-nm imaging case where $NA = 0.93$ and $\sigma = 0.7$. The swing curve period for normally incident light would be 56 nm, but for the partially coherent light it grows slightly to 59 nm. The real

impact is in the reduction of the swing ratio, which for a 300-nm-thick resist would produce $\sin(c)/c = 0.77$, meaning a 23% reduction in the swing ratio.

4.2.5 Effective Absorption

As we shall see, the optimum resist absorption is a strong function of the reflectivity of the substrate. Consider first, however, the simple case of a nonreflecting substrate so that light travels only downward through a resist film of thickness D . The absorption of light through the resist leads to an exposure dose error: a smaller dose at the bottom of the resist (E_{bottom}) compared to the top (E_{top}). The fraction of light making it to the bottom is given by

$$\frac{E_{\text{bottom}}}{E_{\text{top}}} = T_D = e^{-\alpha D} \quad (4.71)$$

As an example, for a resist with $\alpha = 0.5 \mu\text{m}^{-1}$, a 500-nm-thick film will absorb 22% of the light so that $T_D = 0.78$.

If the resist film is coated on a reflective substrate, reflected light traveling up through the film will also be absorbed. The reflected beam will be brighter at the bottom of the resist, so that the sum of the incident and reflected beams will have a smaller variation in dose from top to bottom than for the nonreflective substrate case. The amount can be quantified using one of the previously derived expressions for the standing wave intensity. The ‘average’ intensity (averaged over one standing wave period) was previously derived in Equation (4.50), and is given here normalized so that the intensity at the top of the resist is 1:

$$I_{\text{avg}}(z) = e^{-\alpha z} \left(\frac{1 + |\rho_{23}|^2 e^{-\alpha 2D} e^{\alpha 2z}}{1 + |\rho_{23}|^2 e^{-\alpha 2D}} \right) \quad (4.72)$$

The term $|\rho_{23}|^2 e^{-\alpha 2D}$ represents the fraction of the light intensity that makes it back to the top of the resist after traveling down through the resist, reflecting off the substrate, and traveling back up to the top. It can be thought of as a ‘round-trip’ transmittance and is an important factor in determining the difference between Equation (4.72) and simple bulk absorption.

For small amounts of absorption ($\alpha 2D < 1$, for example), the z -dependent exponential term in the parentheses of Equation (4.72) can be expanded as Taylor series and an approximate expression for the bulk effect can be derived:

$$I_{\text{average}}(z) \approx e^{-\alpha_{\text{eff}} z} \quad (4.73)$$

where the effective absorption coefficient is given by

$$\alpha_{\text{eff}} = \alpha \left(\frac{1 - |\rho_{23}|^2 e^{-\alpha 2D}}{1 + |\rho_{23}|^2 e^{-\alpha 2D}} \right) \quad (4.74)$$

As discussed above, a more reflective substrate actually reduces the bulk intensity variation through the resist, which is expressed here as a lower effective absorption coefficient.

Because of the Taylor series expansion, the expression (4.74) used in Equation (4.73) provides a very good match for the actual intensity as a function of depth near the top of the resist. However, the approximate expression overestimates the amount of absorption, resulting in an intensity that is too low everywhere, with the greatest error at the bottom of the resist. A second approach toward defining an effective absorption is to match the intensity at the top and the bottom of the resist.⁸ This results in an effective absorption coefficient of

$$\alpha_{\text{eff}} = \alpha - \frac{1}{D} \ln \left(\frac{1 + |\rho_{23}|^2}{1 + |\rho_{23}|^2 e^{-\alpha 2D}} \right) \quad (4.75)$$

For the case of small substrate reflectivity (when using a good bottom antireflection coating, for example), this expression simplifies to

$$\alpha_{\text{eff}} \approx \alpha \left[1 - 2|\rho_{23}|^2 \left(\frac{1 - e^{-\alpha 2D}}{\alpha 2D} \right) \right] \quad (4.76)$$

For moderate to small amounts of absorption, this equation can be further simplified to

$$\alpha_{\text{eff}} \approx \alpha [1 - 2|\rho_{23}|^2 (1 - \alpha D)] \quad (4.77)$$

While the intensity using the effective absorption of Equation (4.75) matches the actual average intensity at the top and the bottom of the resist, it gives an intensity that is higher than the actual intensity everywhere else. To match the integrated intensity through the thickness of the resist, the effective absorption coefficient must satisfy the following equation:

$$\frac{\alpha_{\text{eff}}}{1 - e^{-\alpha_{\text{eff}} D}} = \alpha \frac{1 + |\rho_{23}|^2 e^{-\alpha 2D}}{1 - |\rho_{23}|^2 e^{-\alpha 2D} - (1 - |\rho_{23}|^2) e^{-\alpha D}} \quad (4.78)$$

The effective absorption coefficient that matches the integrated intensity will generally lie about halfway between the values given by Equations (4.74) and (4.75).

How can the effective absorption be used when designing resists for different reflectivity applications? One simple design criterion might be to fix the effective absorption coefficient. Suppose that a 0.7- μm -thick deep-UV resist with an absorption coefficient of 0.4 μm^{-1} is currently providing acceptable resist profile results on a bare silicon wafer ($|\rho_{23}|^2 \approx 0.5$). In other words, for the parameters given, the effective absorption provides an acceptable dose variation from the top to the bottom of the resist. From Equation (4.74), the effective absorption coefficient is 0.22 μm^{-1} . Thus, for a resist to have approximately the same profile behavior on a perfectly nonreflecting substrate, its absorption coefficient would have to be lowered to this 0.22- μm^{-1} value. On the other hand, if one wanted to use an equivalent resist on an aluminum substrate with a reflectivity of 0.84 (not necessarily a good idea, given the swing curve effects), one could raise the absorption coefficient to 0.52 μm^{-1} and still exhibit the same effective absorption.

Light traveling at an angle through the photoresist will result in an apparent difference in the absorption coefficient. As will be derived in section 4.8.2, a plane wave traveling through resist at some angle θ_2 will be absorbed according to

$$I(z) \propto \exp(-\alpha z / \cos \theta_2) \quad (4.79)$$

Thus, an effective absorption coefficient can be defined as $\alpha / \cos \theta_2$. If a reflective substrate is present, Equations (4.74)–(4.78) can still be used if $\alpha / \cos \theta_2$ is substituted for α . If a range of angles is used, such as for partially coherent illumination, Equation (4.79) can be integrated over the angular range of the source. For conventional illumination, and for moderate amounts of absorption so that the exponential can be expanded as a Taylor series, the resulting effective absorption coefficient becomes

$$\alpha_{\text{eff}} \approx \alpha \left(\frac{2}{1 + \cos \theta_{2\text{max}}} \right) \quad (4.80)$$

where $\theta_{2\text{max}}$ is the angle in resist corresponding to the maximum angle of the conventional partially coherent source.

4.3 Bottom Antireflection Coatings

As discussed above, reflections from the substrate can cause unwanted variations in the resist profile and swing curve effects. Reflections are caused by a difference in the complex index of refraction of two materials. Since lithography takes place on a variety of substrates with film stacks of many materials and thicknesses, each interface in the film stack can contribute to the overall reflectivity back into the photoresist. This complicated situation is made worse by the inevitable variations in the thicknesses, and sometimes the refractive indices, of the films. One possible solution to reflectivity problems is the bottom antireflection coating (BARC). Also, swing curves can be improved by the use of a top antireflection coating (TARC), described in the next section. These types of antireflection coatings have somewhat different goals, but their basic behavior is the same.

The goals of film stack optimization are to minimize standing waves in the resist, and to reduce the sensitivity of the process to film stack variations (including resist and BARC, but other layers as well). By far the most common way to accomplish all of these goals is by using an optimized BARC. When optimizing a lithography process for reflectivity, there are three basic tasks: (1) optimize the BARC, (2) optimize the resist thickness (from a swing curve perspective) and (3) understand the sensitivity to BARC, resist and film stack variations. For the first task, there are two classes of BARC problems:

- BARC on an absorbing substrate (such as metal) – the goal is to reduce the reflectivity (the thickness of the metal or what is underneath doesn't matter)
- BARC on a transparent substrate (such as silicon dioxide) – reduce the sensitivity to oxide thickness variations (while also keeping reflectivity low)

There is an unfortunate problem in that the substrate reflectivity experienced by the photoresist cannot be measured. Measuring the reflectivity of the substrate when not coated by photoresist is not useful to this task, and once coated the reflectivity becomes hidden from measurement. Thus, the approach that must be used is to calculate the

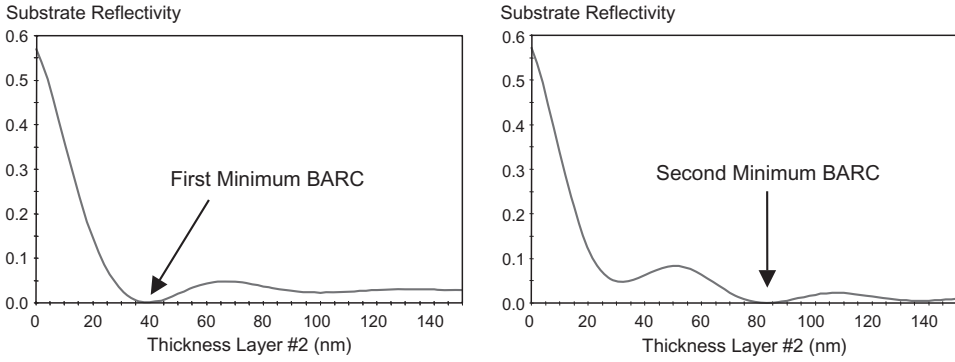


Figure 4.15 Typical examples of substrate reflectivity versus BARC thickness for different resist/BARC/substrate stacks

reflectivity from measured fundamental parameters, namely the thickness and complex refractive index of each layer in the film stack.

For a single-layer BARC, there are three parameters available for optimization: the thickness of the BARC, and the real and imaginary parts of its refractive index. For the simplest use case, a BARC is given and the goal is just to optimize its thickness. Figure 4.15 shows substrate reflectivity calculations for two different resist/BARC/substrate stacks. These reflectivity curves show a characteristic feature of a resist/BARC/substrate stack: the lowest reflectivity could be at the first minimum or at the second minimum depending on the parameters. Etch and process integration considerations determine the range of acceptable BARC thicknesses, and thus the preference for a first-minimum BARC or a second-minimum BARC.

4.3.1 BARC on an Absorbing Substrate

For the case where all of the BARC properties are available for optimization, a more detailed look at the problem is required. Recapping the basic thin film reflectivity theory, the electric field reflection coefficient (the ratio of reflected to incident electric fields) at the interface between two materials is a function of the complex indices of refraction for the two layers. For normal incidence, the reflection coefficient of light traveling through layer i and striking layer j is given by Equation (4.8). For the case of a bottom antireflection coefficient (BARC), assume the BARC (layer 2) is sandwiched between a resist (layer 1) and a very thick substrate (layer 3). The total reflectivity looking down on layer 2 includes reflections from both the top and bottom of the BARC film. The resulting reflectivity, taking into account all possible reflections, is

$$R_{\text{total}} = |\rho_{\text{total}}|^2 = \left| \frac{\rho_{12} + \rho_{23} \tau_D^2}{1 + \rho_{12} \rho_{23} \tau_D^2} \right|^2 \quad (4.81)$$

where the internal transmittance, τ_D , is the change in the electric field as it travels from the top to the bottom of the BARC, given by Equation (4.12).

If the role of layer 2 is to serve as an antireflection coating between materials 1 and 3, one obvious requirement might be to minimize the total reflectivity given by Equation (4.81). If the light reflecting off the top of layer 2 (ρ_{12}) can cancel out the light that travels down through layer 2, reflects off layer 3, and then travels back up through layer 2 ($\rho_{23}\tau_D^2$), then the reflectivity can become exactly zero. In other words,

$$R_{\text{total}} = 0 \quad \text{when} \quad \rho_{12} + \rho_{23}\tau_D^2 = 0 \quad \text{or} \quad \rho_{21} = \rho_{23}\tau_D^2 \quad (4.82)$$

When designing a BARC material, there are only three variables that can be adjusted: the real and imaginary parts of the refractive index of the BARC, and its thickness. One classic solution to Equation (4.82) works perfectly when the materials 1 and 3 are non-absorbing: let $\tau_D^2 = -1$ and $\rho_{12} = \rho_{23}$. This is equivalent to saying that the BARC thickness is a ‘quarter wave’ ($D = \lambda/4n_2$), and the nonabsorbing BARC has a refractive index of $n_2 = \sqrt{n_1 n_3}$. While this BARC solution is ideal for applications like antireflective coatings on lens surfaces, it is not particularly useful for common lithography substrates, which are invariably absorbing.

Will a solution to Equation (4.82) always exist, even when the resist and substrate have complex refractive indices? Since all the terms in Equation (4.82) are complex, zero reflectivity occurs when both the real part and the imaginary part of Equation (4.82) are true. This requirement can be met by adjusting only two of the three BARC parameters (n , κ and D). In other words, there is not just one solution but a family of solutions to the optimum BARC problem. Expressing each reflection coefficient in terms of magnitude and phase,

$$\rho_{ij} = |\rho_{ij}|e^{i\theta_{ij}} \quad (4.83)$$

Equation (4.82) can be expressed as two equalities:

$$D = \frac{\lambda}{4\pi\kappa_2} \ln \left| \frac{\rho_{23}}{\rho_{21}} \right| = \frac{\lambda}{4\pi n_2} (\theta_{21} - \theta_{23}) \quad (4.84)$$

Unfortunately, the seemingly simple forms of Equations (4.82) and (4.84) are deceptive: solving for the unknown complex refractive index of the BARC is exceedingly messy. As a consequence, numerical solutions to Equation (4.82) or (4.84) are almost always used. Note that a second-minimum BARC can be optimized using Equation (4.84) by adding 2π to the angle difference on the right-hand side of the equation.

Consider a common case of a BARC for 193-nm exposure of resist on silicon. The ideal BARC is the family of solutions as shown in Figure (4.16), which gives the ideal BARC n and κ values as a function of BARC thickness. Each solution produces exactly zero reflectivity for normally incident monochromatic light. Within this family of solutions, available materials and the acceptable range of BARC thicknesses (usually constrained by coating and etch requirements) will dictate the final solution chosen.

Another criterion for choosing the optimum BARC is the sensitivity to BARC thickness variations. Often, BARC layers are coated over topography and are partially planarizing. This means that BARC thickness variations across the device are inevitable. Is one BARC solution from the family of solutions given in Figure 4.16 better from a BARC thickness

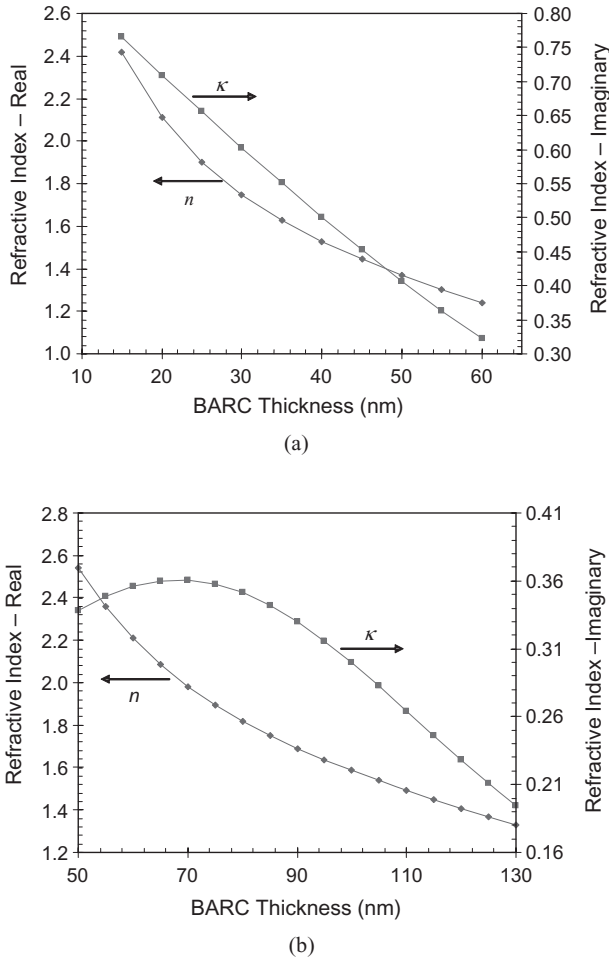


Figure 4.16 Optimum BARC refractive index (real and imaginary parts, n and κ) as a function of BARC thickness for normal incidence illumination (resist index = $1.7 + i0.01536$ and silicon substrate index = $0.8831 + i2.778$) at 193 nm. (a) First minimum BARCs, and (b) second minimum BARCs

sensitivity perspective? Consider only small errors in BARC thickness about the optimum value, so that $D = D_{\text{opt}} + \Delta$ where D_{opt} is the optimum thickness given by Equation (4.84). Since Equation (4.82) is true at the optimum thickness,

$$\rho_{12} + \rho_{23}\tau_D^2 = \rho_{12} + \rho_{23}\tau_{D_{\text{opt}}}^2 e^{i4\pi n_2 \Delta/\lambda} = \rho_{12}(1 - e^{i4\pi n_2 \Delta/\lambda}) \quad (4.85)$$

Using a similar operation on the denominator of the reflectivity expression (4.81) gives

$$R = \left| \frac{\rho_{12}(1 - e^{i4\pi n_2 \Delta/\lambda})}{1 - \rho_{12}^2 e^{i4\pi n_2 \Delta/\lambda}} \right|^2 \quad (4.86)$$

Assuming Δ is small, keeping only the first two terms of a Taylor's expansion of the exponential in the numerator gives

$$R \approx \Delta^2 \left(\frac{\pi n_1}{\lambda} \right)^2 \left| 1 - \left(\frac{n_2}{n_1} \right)^2 \right|^2 \quad (4.87)$$

As can be seen, the reflectivity increases approximately quadratically about the optimum BARC thickness. Different BARC solutions will have different sensitivities, depending on how close the ratio n_2/n_1 is to one. For the first minimum resist on BARC on silicon example above, the ~50-nm BARC thickness solution has the minimum sensitivity to BARC thickness errors.

Similarly, variations in refractive index of a BARC can cause the reflectivity of an otherwise optimal BARC solution to increase. For the case of the optimal normal-incidence BARC, the reflectivity for a given change in BARC index Δn (which can be an error in real and/or imaginary parts) will be approximately

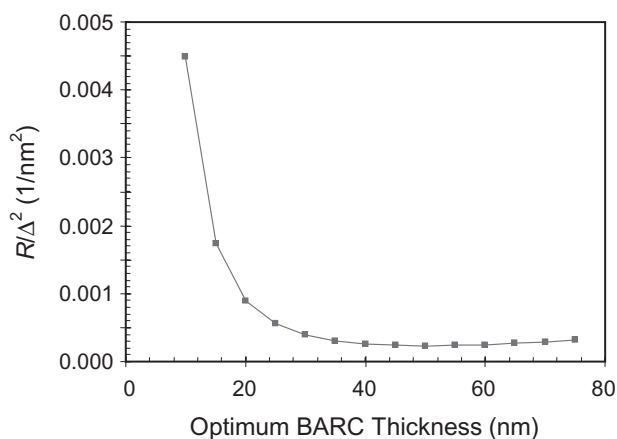
$$R \approx |\Delta n|^2 \left| \frac{\rho_{12}}{1 - \rho_{12}^2} \right|^2 \left| \frac{1 + \rho_{12}}{n_1 - n_2} + \frac{1 - \rho_{23}}{n_2 - n_3} + i4\pi D/\lambda \right|^2 \quad (4.88)$$

Note that $|\Delta n|^2$ is the sum of the squares of the errors in the real and imaginary parts of the index. The sensitivity to BARC errors, either in thickness or in refractive index, is shown in Figures 4.17 and 4.18 for the case of the BARC solutions of Figure 4.16. Note that for the first minimum case, practical BARC solutions have thicknesses in the 20- to 50-nm range. Thus, for this case, thicker BARCs are less sensitive to thickness errors but more sensitive to refractive index errors. Assuming that a reflectivity of 0.1 % can be tolerated, a 20-nm optimal BARC can tolerate 1 nm of thickness error, or a 0.053 change in refractive index. A 40-nm optimal BARC can tolerate 2 nm of thickness error, or a 0.04 change in refractive index. Roughly, if the BARC thickness has no error, the real part of the refractive index of these BARCs must be controlled to 2.5 % assuming no error in the imaginary part, or the imaginary parts must be controlled to 8 % for no error in the real part. Likewise, if the refractive index has no error, the BARC thickness must be controlled to roughly 5 % for the given 0.1 % reflectivity tolerance.

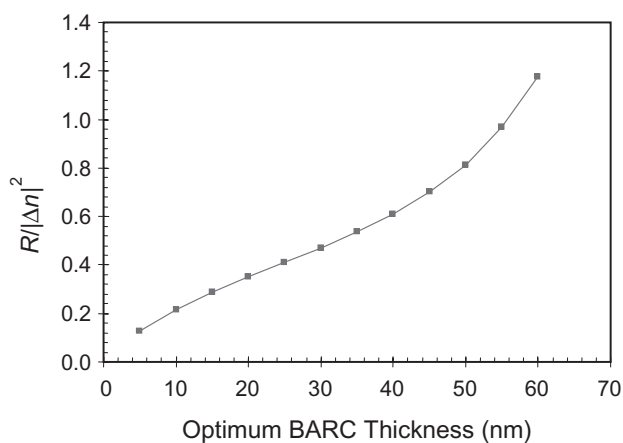
4.3.2 BARCs at High Numerical Apertures

Things become a bit more complicated for the more general case of light traveling at an angle with respect to the film stack normal. In lithographic terms, high numerical apertures allow large ranges of angles to pass through the lens and arrive at the wafer. Off-axis illumination of small pitch patterns produces images made of light concentrated at large angles. Small isolated features with large σ partially coherent illumination create light reaching the wafer over a wide range of angles. And of course, low numerical apertures result in limited ranges of angles reaching the wafer.

How does non-normal incidence affect Equation (4.81)? Each reflection coefficient ρ_{ij} is a function of the angle of light, and a function of polarization. Randomly polarized light (often called unpolarized light, the kind most commonly employed in lithographic tools) can be considered the incoherent sum of two linear and orthogonal polarizations. If θ_i is the incident (and reflected) angle inside the resist and θ_j is the transmitted angle

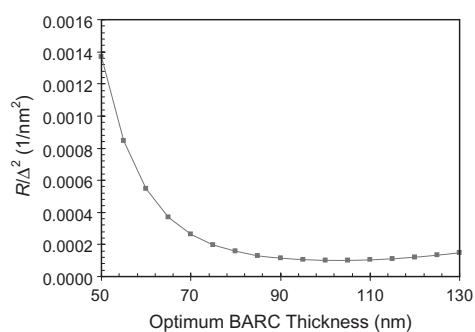


(a)

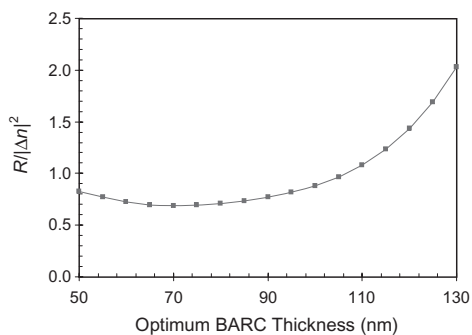


(b)

Figure 4.17 Sensitivity of substrate reflectivity for the optimum first minimum BARCs of Figure 4.16a as a function of (a) BARC thickness errors, or (b) BARC refractive index errors



(a)



(b)

Figure 4.18 Sensitivity of substrate reflectivity for the optimum second minimum BARCs of Figure 4.16b as a function of (a) BARC thickness errors, or (b) BARC refractive index errors

in the BARC, then the electric field reflection and transmission coefficients are given by the Fresnel formulae [Equation (4.29)]. Of course, the relationship between incident and transmitted angle is given by Snell's law, Equation (4.30). The internal transmittance is also a function of angle as given by Equation (4.33).

The requirements for zero reflectivity remain the same. Equation (4.82) must be satisfied for both s - and p -polarization. Since each equation has both real and imaginary parts, there are four constraints that must be satisfied in order to achieve exactly zero reflectivity for unpolarized illumination. However, the BARC film gives us only three degrees of freedom (n , κ and D). In general, there can be no single BARC film that results in zero reflectivity for a non-normal incident randomly polarized plane wave. Calculating the unpolarized intensity reflectivity (defined as the average of the individual reflectivities for s - and p -polarization), the best case optimum BARC parameters are given in Figure 4.19 as a function of incident angle in air (before striking the resist).

Figure 4.20 shows the lowest possible reflectivity as a function of incident angle, using the optimum BARC parameters defined in Figure 4.19 for each angle. As can be seen, the best case reflectivity grows rapidly as the angle increases, and is worse for the thinner BARC film.

Polarization plays a large role in determining the effect of angle on reflectivity. Figures 4.19 and 4.20 show results assuming unpolarized light striking the BARC, while Figure 4.21 separates out the behavior of reflectivity by polarization for two example BARCs. The trend toward using polarized illumination will greatly improve the performance of BARCs at high numerical apertures since only the reflectivity for that one polarization need be optimized.

Stringent CD control requirements demand reflectivities below 0.1%, making BARC design difficult at high numerical apertures with extreme off-axis illumination (i.e. at large incident angles). One potential solution, though not pleasant from a cost and complexity perspective, is to increase the number of free variables available for optimization by using a two-layer or three-layer BARC. This technique, a standard practice in antireflective

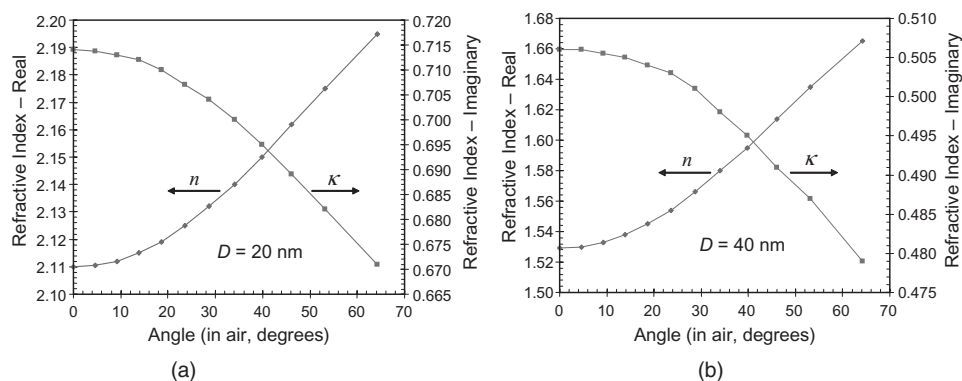


Figure 4.19 Optimum BARC parameters to achieve minimum substrate reflectivity as a function of incident angle (angle defined in air, before entering the photoresist) for two different BARC thicknesses (resist index = $1.7 + i0.01536$ and silicon substrate index = $0.8831 + i2.778$) at 193-nm exposure: (a) 20-nm BARC thickness, and (b) 40-nm BARC thickness

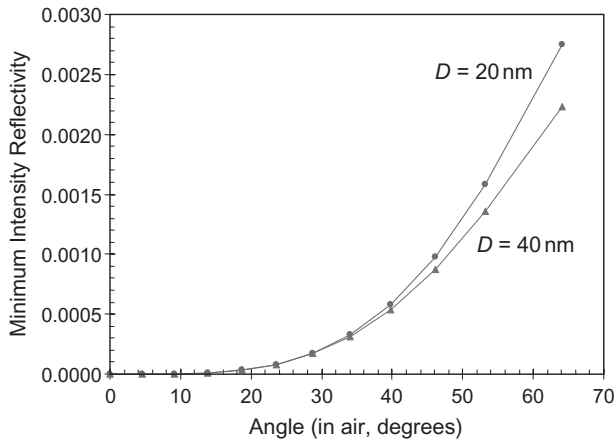


Figure 4.20 The best case (minimum) reflectivity (using the BARC parameters shown in Figure 4.19) of the substrate as a function of incident angle for 20- and 40-nm-thick BARC films. Note that 60° corresponds to the maximum angle in air allowed for $NA = 0.866$

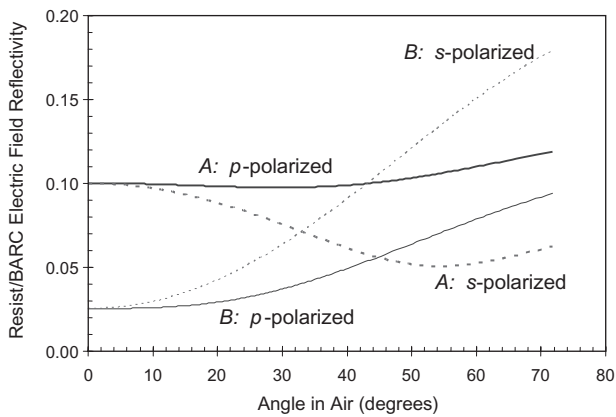


Figure 4.21 An example of the variation of BARC reflectivity as a function of light angle and polarization for two different BARCs. The intensity reflectivity is the square of the electric field reflectivity plotted here, but interference makes the field reflectivity a better measure of the standing wave effects (resist index = $1.7 + i0.01536$, silicon substrate index = $0.8831 + i2.778$, BARC A index = $1.80 + i0.48$, BARC A thickness = 30 nm, BARC B index = $1.53 + i0.54$, BARC B thickness = 39 nm)

coatings for lenses, would provide enough adjustable parameters to make the reflectivity go to zero at normal incidence and at one or more angles, thus providing low reflectivity over a wide range of angles. This approach has become necessary with the advent of immersion lithography and numerical apertures greater than 1.0. Alternately, a gradient index material, with low κ near the top of the BARC gradually increasing toward the bottom, can be used.

4.3.3 BARC on a Transparent Substrate

The second type of BARC optimization problem involves the use of a BARC on a transparent substrate, such as an oxide film. For such a case, the overall substrate reflectivity will be a function of the oxide thickness and one of the goals of the BARC design is to reduce the sensitivity to underlying film thickness variations. As can be seen from Figure 4.22, using the BARC at a first minimum results in a very large sensitivity to underlying oxide thickness variations. In fact, the thickest BARC films provide the most robust behavior when a wide range of oxide thicknesses are expected. Thus, the preferred design approach for this case is to determine the maximum allowed BARC thickness from an integration perspective, then optimize the n and κ values of the BARC to minimize the maximum reflectivity over the range of expected oxide thicknesses.

If the oxide thickness under the BARC varies by only a small amount, the use of a first-minimum BARC becomes practical. The sensitivity to small amounts of underlying oxide thickness variation Δ_{ox} can be derived in a manner similar to Equation (4.87):

$$R \approx \Delta_{\text{ox}}^2 \left(\frac{\pi n_{\text{ox}}}{\lambda} \right)^2 \left| \frac{n_1}{n_2} - \frac{n_2}{n_1} \right|^2 \quad (4.89)$$

where n_{ox} is the oxide refractive index. Of course, while oxide was used here as an example, this analysis applies equally well to any reasonably transparent film under the BARC.

A ‘thick’ BARC solution can be thought of as a BARC that completely absorbs all of the light that may reflect off the substrate and is thus insensitive to changes in the substrate stack. For this case, the product of BARC thickness and the imaginary part of its

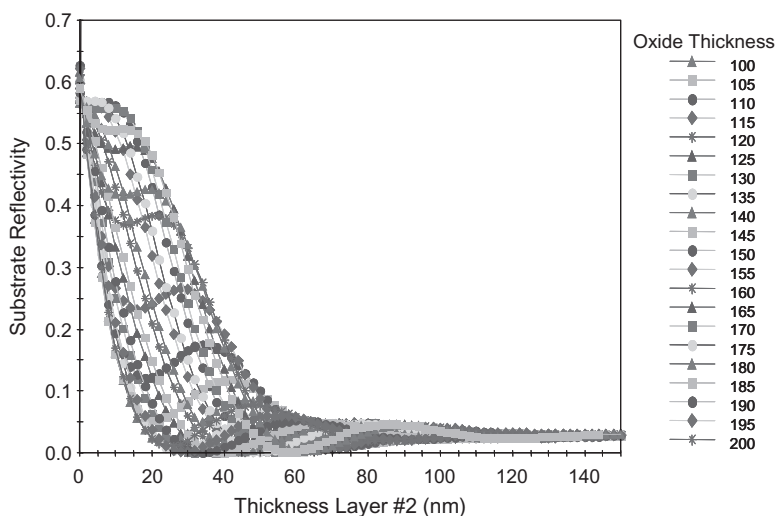


Figure 4.22 Substrate reflectivity versus BARC thickness over a range of underlying oxide thicknesses (oxide on top of a silicon substrate)

refractive index (κ_2) must be sufficiently high to make $\tau_D^2 \approx 0$. Maximum BARC thickness is usually determined by process considerations, thus fixing the minimum κ_2 . Assuming the imaginary part of the resist refractive index is sufficiently small, the resulting reflectivity is

$$R_{\text{total}} = |\rho_{12}|^2 = \left| \frac{n_1 - n_2 - i\kappa_2}{n_1 + n_2 + i\kappa_2} \right|^2 = \frac{(n_1 - n_2)^2 + \kappa_2^2}{(n_1 + n_2)^2 + \kappa_2^2} \quad (4.90)$$

The optimum real part of the BARC refractive index to minimize reflectivity will then be

$$n_2 = \sqrt{n_1^2 + \kappa_2^2} \quad (4.91)$$

The resulting reflectivity is

$$R = \frac{n_2 - n_1}{n_2 + n_1} \approx \frac{\kappa_2^2}{4n_1^2 + \kappa_2^2} \quad (4.92)$$

Obviously, the thick BARC solution is insensitive to BARC thickness variations by design. The sensitivity to errors in BARC refractive index can be obtained from Equation (4.90). For a small change in the real part of the BARC refractive index,

$$\frac{\Delta R}{R} \approx \Delta n_2^2 \frac{n_1}{n_2(n_2^2 - n_1^2)} = \Delta n_2^2 \frac{n_1}{n_2 \kappa_2^2} \quad (4.93)$$

For a small change in the imaginary part of the BARC refractive index,

$$\frac{\Delta R}{R} \approx \frac{\Delta \kappa_2}{\kappa_2} \left(\frac{2n_1}{n_2} \right) \quad (4.94)$$

4.3.4 BARC Performance

How critical is BARC optimization? How low must the substrate reflectivity be before acceptable CD control can be expected? There is no single answer to these questions, since they are process and feature dependent. But consider the example shown in Figure 4.23. Here, 100-nm lines on a 280-nm pitch are simulated with a stepper using annular illumination, with a center sigma given by $\sigma NA = 0.54$. As can be seen, a substrate reflectivity of less than 0.1 % still leads to a noticeable swing behavior.

A common approach to characterizing the impact of substrate reflectance (that is, the impact of a nonideal BARC) is to relate reflectivity in the presence of film thickness variations to effective dose errors. Consider, for example, the case where topography on the wafer results in resist thickness variations that extend beyond one swing curve period. For such a case, the swing ratio is effectively the fractional dose error range resulting from resist thickness variations.

$$\frac{\Delta E}{E} \approx SR \approx \alpha \Delta D + 4|\rho_{12}\rho_{23}|e^{-\alpha D} \quad (4.95)$$

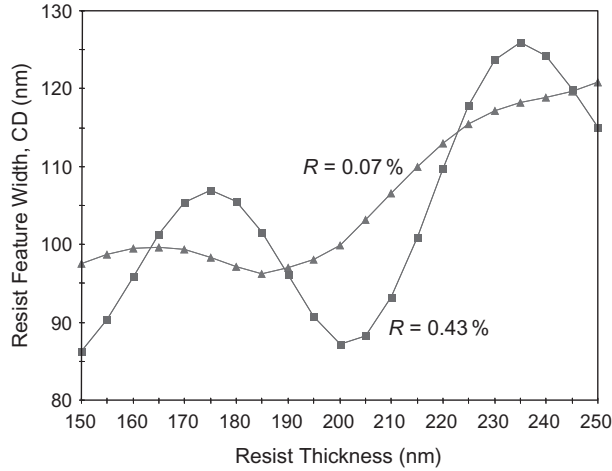


Figure 4.23 CD swing curves (100-nm lines with a 280-nm pitch are printed with a stepper using annular illumination, with a center sigma given by $\sigma\text{NA} = 0.54$) for two different BARCs with different levels of optimization, as given by the resulting substrate reflectivity R

Consider a 193-nm resist with $\alpha = 1.2 \mu\text{m}^{-1}$, $D = 200 \text{ nm}$, a resist thickness variation equal to the swing period of 56 nm, and $|\rho_{12}| = 0.26$. This results in a fractional effective dose error of

$$\frac{\Delta E}{E} \approx 0.067 + 0.83|\rho_{23}| \quad (4.96)$$

For a BARC that allows 1% reflectance ($|\rho_{23}| = 0.1$), bulk absorbance adds 6.7% effective dose error while the swing effect adds another 8.3% dose error. Both of these numbers are clearly too high, so that a 193-nm lithography process must employ planarization to reduce the resist thickness variations caused by topography.

For a 248-nm resist with $\alpha = 0.5 \mu\text{m}^{-1}$, $D = 400 \text{ nm}$, a resist thickness variation equal to the swing period of 70 nm, and $|\rho_{12}| = 0.275$, the fractional effective dose error is

$$\frac{\Delta E}{E} \approx 0.035 + 0.90|\rho_{23}| \quad (4.97)$$

For a BARC that allows 1% reflectance ($|\rho_{23}| = 0.1$), bulk absorbance adds 3.5% effective dose error while the swing effect adds another 9% dose error.

A more thorough ‘dose budget’ analysis would consider variations in resist thickness, BARC thickness and underlying film stack simultaneously to determine the effective dose errors caused by changes in the energy coupled into the resist. Equation (4.95) assumed sufficient topography to cause one full swing period of resist thickness variations. When smaller amounts of resist thickness variation are present, the effective dose errors are also smaller. Consider a resist thickness $D = D_{\min} \pm \Delta$ where $D_{\min}$ is the thickness at the swing curve minimum and Δ is small compared to the swing period.

Equation (4.51) can be used to derive the effective dose error for a small resist thickness error:

$$\frac{\Delta E}{E} \approx \left(\frac{[e^{\alpha\Delta} - 1] + 2|\rho_{12}\rho_{23}|e^{-\alpha D}[1 - \cos(4\pi n_2\Delta/\lambda)]}{1 - 2|\rho_{12}\rho_{23}|e^{-\alpha D}} \right) \quad (4.98)$$

which can be further approximated for small Δ as

$$\frac{\Delta E}{E} \approx \left(\frac{\alpha\Delta + 4|\rho_{12}\rho_{23}|e^{-\alpha D}(2\pi n_2\Delta/\lambda)^2}{1 - 2|\rho_{12}\rho_{23}|e^{-\alpha D}} \right) \quad (4.99)$$

and for a reasonably good BARC as

$$\frac{\Delta E}{E} \approx \alpha\Delta + 4|\rho_{12}\rho_{23}|e^{-\alpha D}(2\pi n_2\Delta/\lambda)^2 \quad (4.100)$$

The dose error comes from a bulk absorption term ($\alpha\Delta$) plus a fraction of the swing amplitude. Consider the standard 193-nm resist on BARC on silicon case described above, with $\alpha = 1.2\mu\text{m}^{-1}$, $D = 200\text{ nm}$, $n_{\text{resist}} = 1.7$, and $|\rho_{\text{air-resist}}| = 0.26$. For Δ in nanometers,

$$\frac{\Delta E}{E} \approx 0.0012\Delta + 0.0025|\rho_{23}|\Delta^2 \quad (4.101)$$

For an 8.3-nm increase in resist thickness, the bulk effect causes about a 1% effective dose error. The swing effect will cause another 1% effective dose error for that 8.3-nm resist thickness error if the substrate reflectivity is 0.33%. The maximum allowed substrate reflectivity is a function of the range of resist thickness that must be tolerated and the amount of effective dose error one is willing to accept. Specifications of substrate reflectivity in the 0.1–0.5% range are not uncommon.

Swing curves are extremely important in lithography for one simple reason: topography on the wafer can lead to variations in resist thickness on the order of a period in the swing curve or more. In some processes, these effects are the leading cause of CD errors on some critical mask levels. Figure 4.24 shows an example of how a line printing over a topographical step on the wafer can lead to linewidth variations due to swing curve effects.

BARCs of course can dramatically reduce the linewidth variation caused by topography on the wafer. A second problem that BARCs can solve is *reflective notching*. As Figure 4.25 illustrates, light can reflect off the tilted edges of reflective topography. Patterns being printed near such a reflective edge may receive extra exposure from the reflection, causing a notch, usually near the top of the feature. BARCs can also significantly suppress reflective notching (Figure 4.26).

4.4 Top Antireflection Coatings

As we saw in Equation (4.46), the amplitude of the reflectivity swing curve is determined by the multiplier of the cosine term, $2|\rho_{12}\rho_{23}|e^{-\alpha D}$. It is clear that this term can be reduced in three ways: (1) increase absorption (increase the term αD) through the use of a dyed

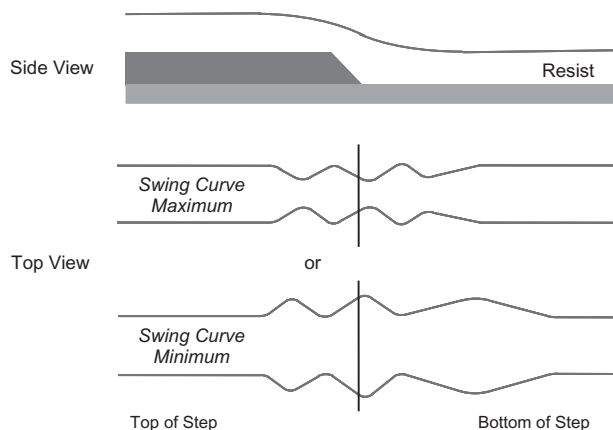


Figure 4.24 Example of how resist thickness variations over topography produce linewidth variations due to swing curve effects when a BARC is not used

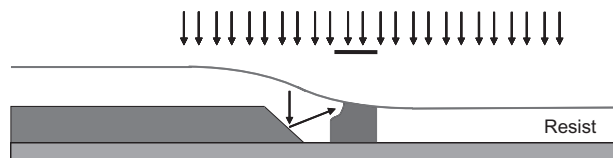


Figure 4.25 Reflective notching occurs when nearby topography reflects light obliquely into an adjacent photoresist feature

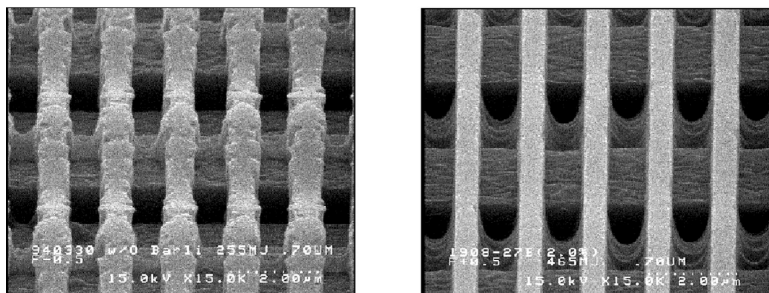


Figure 4.26 Imaging of lines and spaces over reflective topography without BARC (left) showing reflective notching, and with BARC (right) showing the reflective notching effectively suppressed (photos courtesy of AZ Photoresist, used with permission)

resist; (2) decrease substrate reflectivity (decrease ρ_{23}) by using a BARC; and (3) decrease resist reflectivity (decrease ρ_{12}) by using a top antireflection coating (TARC).^{9,10} Looking again at Figure 4.12b, the reflectivity swing curve comes about through the interference of E_{r0} with the other reflected rays such as E_{r1} . By eliminating E_{r0} using a TARC, there can be no interference and thus no swing curve.

One can simplify the design of the TARC somewhat by imagining the photoresist as infinitely thick. Thus, Equation (4.81) applies where layer 2 is the TARC and layer 3 is the resist. If the role of layer 2 is to serve as an antireflection coating between materials 1 and 3, one obvious requirement might be to minimize the total reflectivity given by Equation (4.81). If the light reflecting off the top of layer 2 (ρ_{12}) can cancel out the light which travels down through layer 2, reflects off layer 3, and then travels back up through layer 2 ($\rho_{23}\tau_D^2$), then the reflectivity can become exactly zero [in other words, when Equation (4.82) is true for the TARC].

When designing an antireflection coating material, there are only three variables that can be adjusted: the real and imaginary parts of the refractive index of the coating, and its thickness. One classic solution to Equation (4.82) works very well when the materials 1 and 3 are not very absorbing. It is clear that Equation (4.82) is satisfied when $\tau_D^2 = -1$ and $\rho_{12} = \rho_{23}$. The requirement that $\tau_D^2 = -1$ means that two passes of the light through the TARC cause a 180° phase change with no absorption (since the magnitude is still one). From the definition of the internal transmittance, this means that the TARC thickness must be adjusted to a ‘quarter wave’:

$$D = \frac{\lambda}{4n_2} \quad (4.102)$$

The requirement that $\rho_{12} = \rho_{23}$ will be satisfied when the index of refraction of the TARC is made to be

$$n_2 = \sqrt{n_1 n_3} \quad (4.103)$$

Further, since the TARC does not absorb, the imaginary part of its index is zero. Thus, Equation (4.82) can only be true if both materials 1 and 3 have no imaginary parts to their indices of refraction. For a resist with a refractive index of 1.7 at a wavelength of 193 nm (in air), the optimum TARC will have a refractive index of about 1.30 and a thickness of 37 nm.

An antireflection layer defined by Equations (4.102) and (4.103) will have zero reflectivity. Also, due to the lack of absorption, 100% of the light striking the TARC will be transmitted into layer 3. Thus, this type of antireflection coating is commonly used for coating optical components (such as camera or stepper lenses) where the goal is not so much reducing the reflectivity as it is maximizing the transmittance. The ‘maximum transmittance’ type of antireflection coating is also used for top antireflection coatings. This perfect TARC solution, however, is only available for the special case when both layers 1 and 3 are transparent. Thus, since in reality resist will always be somewhat absorbing, a perfect TARC is not possible, with a residual reflectance off of the TARC proportional to the imaginary part of the resist refractive index squared. However, since resists have an imaginary part of the refractive index in the range of 0.01–0.02, the impact of resist absorption on TARC performance is quite negligible.

More problematically, it is very difficult to find practical materials with refractive indices low enough to make a near ideal TARC. Most available TARCs have refractive indices greater than 1.4 (versus an ideal index closer to 1.3). Letting n_{opt} be the optimum TARC refractive index given by Equation (4.103) and assuming a quarter wave thickness

of TARC is always used, the reflectivity resulting from the use of a nonoptimum TARC refractive index is

$$R = \left[\frac{n_{\text{TARC}}^2 - n_{\text{opt}}^2}{n_{\text{opt}}^2 + n_{\text{TARC}}^2} \right]^2 \quad (4.104)$$

If the TARC refractive index is close to the optimum, this reflectivity is approximately

$$R \approx \left[\frac{n_{\text{TARC}} - n_{\text{opt}}}{n_{\text{opt}}} \right]^2 \quad (4.105)$$

Like BARCs, the performance of TARCs is degraded when a range in incident angles of light is used. Both the optimum TARC thickness and the optimum refractive index will be a function of incident angle and polarization.

4.5 Contrast Enhancement Layer

A *contrast enhancement layer* (CEL) is a highly bleachable coating placed on top of the photoresist that serves to enhance the contrast of an aerial image projected through it.¹¹ Unlike a TARC, where the real part of the refractive index is chosen to maximize transmittance and the imaginary part is kept as low as possible, the CEL works by having a high absorbance that ‘bleaches’, i.e. that becomes more transparent as it is exposed. The mechanism of CEL bleaching is similar to the bleaching that occurs in g-line and i-line photoresists, which will be covered in Chapter 5.

The CEL is initially opaque so that essentially no light is transmitted into the photoresist. Clear areas of the mask, and thus areas of high intensity of the aerial image, produce exposure of the CEL, which begins to bleach in these regions. As the CEL bleaches, light is transmitted into the photoresist where it can begin to expose the resist. In the regions of low intensity, however, the CEL remains more opaque and less transmitting. As a result, the image transmitted into the resist exhibits a steeper transition from bright to dark (Figure 4.27).

CELs are not commonly used for submicron semiconductor lithography. Their main drawback, besides the expense of the material and added processing, is the significant increase in exposure dose (in the range of 2–3 times higher) required to bleach the CEL.

4.6 Impact of the Phase of the Substrate Reflectance

The refractive index of a material is complex in general, the imaginary part being directly proportional to the absorption coefficient of the material. Thus, in general, the reflection coefficient between two materials will be a complex number. The magnitude of ρ determines the magnitude of the reflected light while its phase gives a phase change upon reflection. As we’ve seen, it is the magnitude of the reflection coefficient that impacts the amplitudes of standing waves and swing curves. But the phase will have an impact on

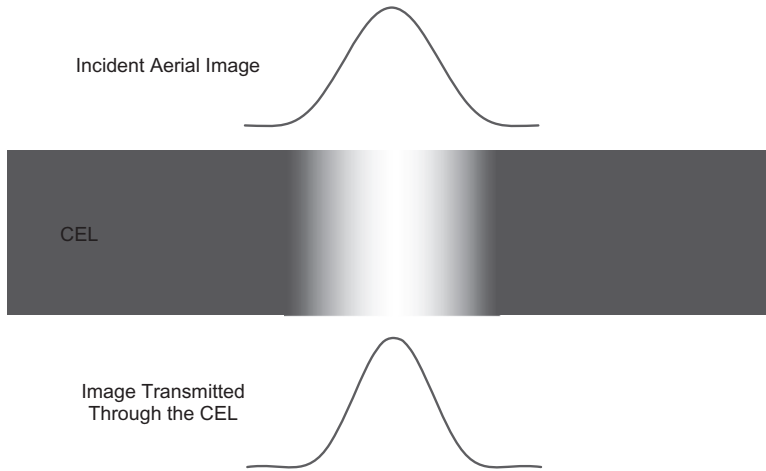


Figure 4.27 Contrast Enhancement Layer (CEL) bleaching improves the quality of the aerial image transmitted into the photoresist

lithography as well. Consider a typical resist on silicon at the i-line wavelength (365 nm). The magnitude of the reflection coefficient is about 0.63 with a phase of -169° . For resist on aluminum, $\rho = 0.93 \angle -138^\circ$ (the notation meaning that the reflection coefficient has a magnitude of 0.93 and a phase angle of -138°). For more complicated layered substrates, both the magnitude and phase of the reflection coefficient will depend on the thicknesses of the various layers.

Consider a simple but common film stack: resist (1 μm thick) on silicon nitride (100 nm) on silicon dioxide (40 nm) on silicon. The reflection coefficient between the resist and the underlying film stack is a function of the optical properties of all of the materials, but also of the thickness of the nitride and the oxide films. For example, variation of the nitride thickness leads to a moderate variation in the magnitude and a large variation in the phase of the reflection coefficient, as shown in Figure 4.28. Oxide thickness variations produce similar effects.

What will be the lithographic effects of this nitride thickness variation? The $\pm 7\%$ change in the magnitude of the reflection coefficient will have some subtle effects, but the large change in the phase of the reflection will cause two major problems. The resist swing curve is a function of the phase change of light that passes down and back up through the resist. Changes in resist thickness cause a change in this phase, giving rise to a sinusoidal variation in dose-to-clear (E_0) and linewidth. Any change in the phase upon reflection will produce the same effect [a variation in ϕ_{23} in Equation (4.51), for example]. Figure 4.29 shows two resist swing curves corresponding to two nitride thicknesses (minimum and maximum thicknesses from Figure 4.28). The nitride thickness variation produces its own swing curve, resulting in linewidth variations of the same magnitude as for resist thickness variations. Control of the nitride thickness (and oxide thickness, for that matter) is just as critical as resist thickness control.

The second effect of this phase change upon reflection is on the shape of the resulting resist profile. When the reflected light is 180° out of phase with the incident light,

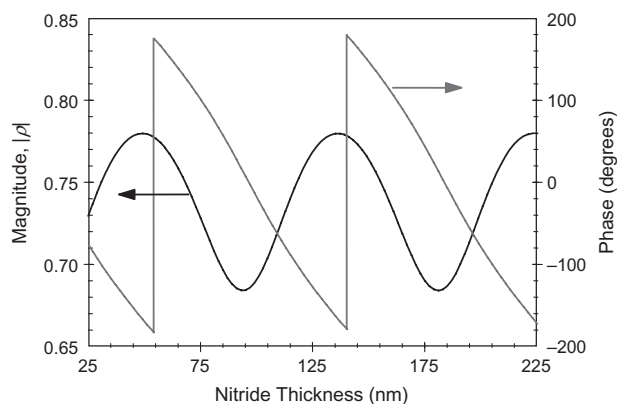


Figure 4.28 Variation of the magnitude and phase of the resist/substrate reflection coefficient as a function of silicon nitride thickness for a film stack of resist on nitride on 40 nm of oxide on silicon

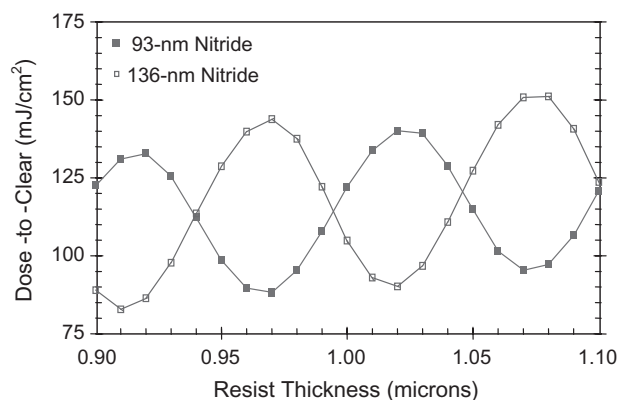


Figure 4.29 Changes in nitride thickness cause a shift in the phase of the resist swing curve, making nitride thickness control as critical as resist thickness control

destructive interference results in a minimum light intensity at the resist/substrate interface. When the reflected light is in phase with the incident light, constructive interference results in a maximum light intensity at the resist/substrate interface. Although standing waves are generally smoothed out by post-exposure bake diffusion, any asymmetry in the standing wave pattern inside the resist can lead to less than perfect reduction of the amplitude. When the phase change upon reflection is about $+90^\circ$ or -90° , the local region near the interface has an average intensity that is higher or lower, respectively, than the average through the bulk. The result is resist undercutting and resist footing. Figure 4.30 shows typical resist profile shapes at different nitride thicknesses.

One can see that nitride thickness variations of just ± 20 nm can have huge effects on both linewidth and resist profile shape. If oxide thickness can vary as well, the requirements for nitride thickness control become even tighter. In many cases, the

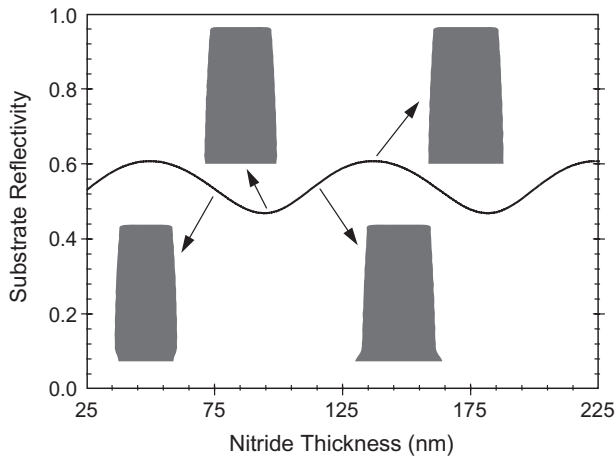


Figure 4.30 Nitride thickness also affects the shape of the resist profile, causing resist footing, undercuts or vertical profiles. Substrate reflectivity (the square of the magnitude of the reflection coefficient) is shown for comparison

lithographic requirements for thin film thickness uniformity and control far exceed other device-related restrictions on these thicknesses. As with other swing curve effects, a BARC can be very effective at reducing the sensitivity to underlying film stack variations.

4.7 Imaging in Resist

The aerial image is, quite literally, the image in air. In the world of semiconductor lithography, it is the image of a photomask projected onto the plane of the wafer, but assuming that only air occupies this space rather than the resist-coated wafer. Although aerial images do not really exist in lithography, the aerial image, which is reasonably easy to calculate though very difficult to measure, is a convenient proxy for the final resist image. By picking an exposure-dose dependent intensity threshold, an estimate of the final resist CD is obtained (this essentially assumes that the resist is ideal, with an infinite contrast). Such uses are based on the idea that an aerial image is a good predictor of what the resist image will look like. However, at high numerical apertures the aerial image is in fact a very poor predictor of the final resist image. Due to vector effects, the image in resist can be greatly different from the aerial image, as will be discussed below.

4.7.1 Image in Resist Contrast

At high numerical apertures, the vector nature of light affects the formation of an image as a function of the polarization of the light and the angles of the various diffraction orders that add together to form the image (see Chapter 3). Two plane waves, approaching a wafer at different angles, will interfere to form fringe patterns of light and dark making the simplest of lithographic patterns – an array of lines and spaces. This phenomenon of

interference is the key to pattern formation. Without it, light hitting the wafer from different directions would simply add to give a uniform intensity. We can see mathematically the effect of interference by examining how two electric fields combine to form a resultant electric field (magnitude and intensity). Ignoring a few details that will be described in the following section, the intensity of light I is the square of the magnitude of the electric field E . If two electric fields are combined, what is the intensity of the combination? If the two electric fields do not interfere, the total intensity is the sum of the individual intensities (i.e. it is a constant).

$$I = |E_1|^2 + |E_2|^2 \quad (4.106)$$

If, however, the two electric fields interfere completely, the total intensity will be

$$I = |E_1 + E_2|^2 \quad (4.107)$$

Here, the variation in the phase of E_1 relative to E_2 produces a spatial variation in intensity that is our image.

As discussed in detail in Chapters 2 and 3, two electric fields interfere only if two conditions are met: (1) there is a fixed phase relationship between the electric fields, and (2) there is some overlap in electric field direction. The first condition is met when the electric fields arise from diffraction orders created from a single source point. The second condition means that the two electric fields interfere with each other only to the extent that their electric fields oscillate in the same direction. If the electric fields are at right angles to each other, there will be no interference. Thus, to determine the amount of interference between two electric fields, one must first determine the amount of directional overlap between them. Standard vector mathematics gives us some simple tools to calculate directional overlap and thus the amount of interference. When thinking of light as a scalar rather than a vector quantity, we ignore the subtleties discussed above. Essentially, a scalar description of light always assumes that the electric fields are 100% overlapped and all electric fields add together as in Equation (4.107). Since interference is what gives us the patterns we want, a scalar view of light is generally too optimistic. The vector description of light says that there is usually some fraction of the electric fields of our two vectors that don't overlap and thus don't contribute to interference. The noninterfering light is still there, but it adds as a uniform intensity that degrades the quality of the image.

As was done in Chapter 3, consider the interference of two plane waves approaching the wafer at fairly large angles (Figure 4.31). The electric field can point in any direction perpendicular to the direction of propagation. This arbitrary direction can in turn be expressed as the sum of any two orthogonal (basis) directions. The most convenient basis directions are called transverse electric or TE (the electric field pointing out of the page of the drawing) and transverse magnetic or TM (the electric field pointing in the page of the drawing). The TE case means that the electric fields of the two plane waves are always 100% overlapped regardless of the angle between the plane waves. For the TM case, however, the extent of overlap between the two vectors grows smaller as the angle between the plane waves grows larger.

We can calculate how much the two electric fields for the TM case will interfere with each other. Suppose that two rays are traveling at an angle θ with respect to the vertical direction (i.e. the direction normal to the wafer), as shown in Figure 4.31. The electric

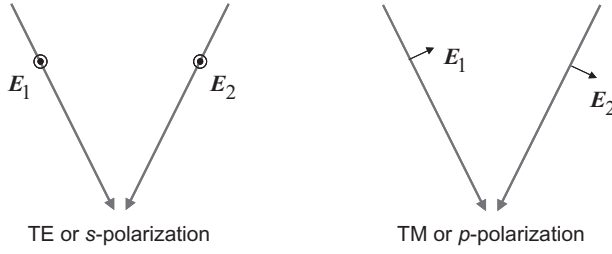


Figure 4.31 Two plane waves with different polarizations will interfere very differently. For transverse electric (TE) polarization (electric field vectors pointing out of the page), the electric fields of the two vectors overlap completely regardless of the angle between the interfering beams

fields E_1 and E_2 will have an angle between them of 2θ . The amount of the electric field vector E_2 that points in the same direction as E_1 is just the geometric projection, $E_2 \cos(2\theta)$. Thus, the intensity will be given by the coherent (electric field) sum of the parts that overlap plus the incoherent (intensity) sum of the parts that don't overlap.

$$I = |E_1 + E_2 \cos(2\theta)|^2 + |E_2 \sin(2\theta)|^2 \quad (4.108)$$

Note that for $\theta = 0$, this equation reverts to the perfectly coherent (interfering) sum of Equation (4.107). For $\theta = 45^\circ$, the two electric fields are perpendicular to each other and Equation (4.108) becomes the perfectly incoherent (noninterfering) sum of Equation (4.106). If we consider the simplest case of two unit amplitude plane waves,

$$\begin{aligned} E_1 &= e^{i2\pi(z \cos \theta + x \sin \theta)/\lambda}, E_2 = e^{i2\pi(z \cos \theta - x \sin \theta)/\lambda} \\ I_{\text{TE}}(x) &= 2 + 2 \cos(4\pi x \sin \theta / \lambda) \\ I_{\text{TM}}(x) &= 2 + 2 \cos(2\theta) \cos(4\pi x \sin \theta / \lambda) \end{aligned} \quad (4.109)$$

Note that for both TE and TM illumination, the resulting images have no z -dependence. In other words, the simple two-beam imaging case has infinite depth of focus.

Two common image metrics, the normalized image log-slope (NILS) and the image contrast will now be calculated for the images in Equation (4.109). For the TE image, the NILS (assuming the desired image is equal lines and spaces) is equal to π , and for the TM image it is $\pi \cos(2\theta)$. The visibility (or contrast) of the resulting fringes is the difference of the maximum and minimum intensities of the interference pattern divided by the sum of these two quantities. For the TE polarization case, the contrast is always exactly 1. For the TM case, the contrast depends on the angle between the two waves and is equal to $\cos(2\theta)$. Thus, as the angle between the plane waves increases, the contrast of the resulting image decreases. One can see why the TM polarization is often called the 'bad' polarization – it provides less interference and reduced image quality.

The resist is not exposed by an aerial image, but by the image in resist. The plane waves that interfere to form the image first propagate into the resist. Once in the resist, they can interfere to form an image in the resist. This image will be different than the aerial image due to refraction. As each plane wave travels from air to resist, refraction lowers the angle of the light according to Snell's law. Letting n be the refractive index of

the resist (and assuming the air above the resist has an index of refraction of 1.0), the angle of one of the plane waves inside the resist will be

$$\sin(\theta_{\text{resist}}) = \frac{1}{n} \sin(\theta_{\text{air}}) \quad (4.110)$$

For the TE light, the interference will be the same and the two plane waves will produce a sinusoidal image of contrast 1. For the TM light, the contrast will be

$$\text{TM Contrast} = \cos(2\theta_{\text{resist}}) = 1 - 2 \sin^2(\theta_{\text{resist}}) = 1 - \frac{2}{n^2} \sin^2(\theta_{\text{air}}) \quad (4.111)$$

Figure 4.32 illustrates the difference between the NILS and the contrast in air versus in resist as a function of angle for TM light when the resist refractive index is 1.7. For an angle of 30° (corresponding to two-beam imaging at the resolution limit of a lens of a modest $\text{NA} = 0.5$), the aerial image for TM illumination has a contrast of 0.5, while the image in resist has a much more acceptable contrast of 0.83. Unpolarized light, which produces an average of the TE and TM images, produces a contrast that is also the average between 1 and the value given by Equation (4.111) (ignoring the difference in transmission of the two polarizations into the resist). Thus,

$$\text{Unpolarized Contrast} \approx 1 - \frac{1}{n^2} \sin^2(\theta_{\text{air}}) \quad (4.112)$$

For the case of unpolarized two-beam imaging at the resolution limit of an $\text{NA} = 0.9$ lens, the aerial image would show a contrast of 0.19 (obviously unacceptable) while the image in resist would have a more reasonable contrast of 0.72. In fact, though, Equation (4.112)

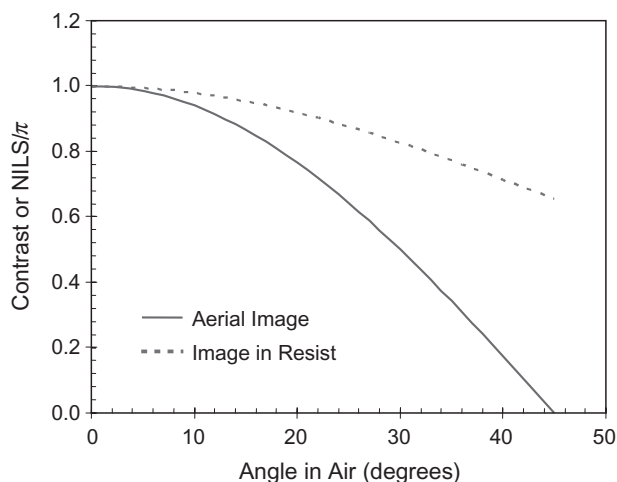


Figure 4.32 The interference between two TM polarized plane waves produces an image whose contrast and NILS depend on the angle. Since the angle in resist is reduced by refraction, the contrast and NILS of the image in resist are better than those of the aerial image

is optimistic since TM light will always transmit into the photoresist better than TE light. Using the data in Figure 4.5, for example, at an incident angle of 60° almost 100 % of the TM light is transmitted into the resist, whereas only 75 % of the high-contrast-producing TE light makes it into the resist.

What does all of this mean? When calculating aerial images, vector effects can dramatically alter the resulting image compared to an approximate scalar calculation. The resist, however, mitigates some of these vector effects and the image in resist can be dramatically different from the aerial image. When trying to approximate a resist feature by a calculated intensity image, only the image in resist (calculated correctly using the vector nature of light) can be expected to give reasonable results.

4.7.2 Calculating the Image in Resist

An aerial image is formed by the interference of plane waves that result from discrete diffraction orders passing through the lens. One of the simplest cases to consider is the imaging of small lines and spaces so that only the zero and the two first orders travel through the lens. For coherent illumination, the zero order will be a plane wave traveling in the z -direction, with magnitude a_0 . The two first orders will be plane waves each with magnitudes a_1 and traveling at angles given by

$$n \sin \theta = \pm \lambda / p \quad (4.113)$$

where p is the pitch of the line/space pattern and n is the refractive index of the media. Ignoring for a moment the reflecting substrate, the image will be formed by the interference (sum) of these three plane waves. Assuming that the image is focused at $z = 0$ (the z -position where all three of the plane waves have the same phase), the resulting electric field of the image will be, for s -polarization,

$$E(x, z) = a_0 e^{ikz} + a_1 e^{ikx \sin \theta} e^{ikz \cos \theta} + a_1 e^{-ikx \sin \theta} e^{ikz \cos \theta} \quad (4.114)$$

This sum can be simplified into a more common and convenient form as

$$E(x, z) = e^{ikz} (a_0 + 2a_1 \cos(2\pi x/p) e^{-ikz(1-\cos \theta)})$$

giving

$$I(x, z) = a_0^2 + 2a_1^2 + 4a_0 a_1 \cos(kz(1-\cos \theta)) \cos(2\pi x/p) + 2a_1^2 \cos(4\pi x/p) \quad (4.115)$$

Equation (4.115) is the standard s -polarized three-beam image where z can be interpreted as the distance from best focus. A similar expression for p -polarized illumination can also be derived:

$$\begin{aligned} E_x &= e^{ikz} (a_0 + 2a_1 \cos \theta \cos(2\pi x/p) e^{-ikz(1-\cos \theta)}) \\ E_z &= ie^{ikz \cos \theta} (2a_1 \sin \theta \sin(2\pi x/p)) \end{aligned} \quad (4.116)$$

giving

$$\begin{aligned} I(x, z) &= a_0^2 + 2a_1^2 + 4a_0 a_1 \cos \theta \cos(kz(1-\cos \theta)) \cos(2\pi x/p) \\ &\quad + 2a_1^2 \cos(2\theta) \cos(4\pi x/p) \end{aligned} \quad (4.117)$$

To determine the aerial image in the presence of a reflecting substrate, the same procedure is followed as above, but including the reflected plane waves as well. The result is, for s-polarization,

$$E(x, z) = e^{ikz} (a_0 [1 + \rho(0) e^{-ik2z}] + 2a_1 \cos(2\pi x/p) e^{-ikz(1-\cos\theta)} [1 + \rho(\theta) e^{-ik2z\cos\theta}]) \quad (4.118)$$

In this equation, interference between plane waves causes a variation of the electric field in the x -direction (the image) and a variation in the z -direction (a combination of defocus and standing waves). In fact, one can think of each plane wave as creating its own electric field standing wave, with the final image as the weighted sum of these standing waves:

$$\begin{aligned} E_{\text{sw}}(\theta, z) &= e^{ikz\cos\theta} + \rho(\theta) e^{-ikz\cos\theta} \\ E(x, z) &= a_0 E_{\text{sw}}(0, z) + 2a_1 \cos(2\pi x/p) E_{\text{sw}}(\theta, z) \end{aligned} \quad (4.119)$$

The above image in resist analysis captured the essence of the physics involved, but ignored the air-resist interface and the multiple reflections that would result. However, the final form of Equation (4.119) suggests that the more detailed standing wave expression of Equation (4.32) can be used as E_{sw} in Equation (4.119) to get an accurate image in resist result. Alternately, the diffraction order amplitudes a_0 and a_1 can be replaced by their effective ‘in-resist’ amplitudes:

$$\begin{aligned} a_{0r} &= a_0 \frac{\tau_{12}(0)}{1 + \rho_{12}(0)\rho_{23}(0)\tau_D^2(0)} \\ a_{1r} &= a_1 \frac{\tau_{12}(\theta)}{1 + \rho_{12}(\theta)\rho_{23}(\theta)\tau_D^2(\theta)} \end{aligned} \quad (4.120)$$

and the simplified standing wave z -dependence would become

$$E_{\text{sw}}(\theta, z) = e^{ikz\cos\theta} + \rho(\theta)\tau_D^2(\theta)e^{-ikz\cos\theta} \quad (4.121)$$

In these two equations, θ now represents the angle in the resist.

Calculating the intensity in the resist from Equation (4.121) results in cumbersome algebra. However, for the case of a low-reflectivity substrate (through the use of a reasonably good BARC, for example), some of the resulting terms can be ignored, giving

$$\begin{aligned} I(x, z) &\approx |a_{0r}|^2 I_{\text{sw}}(0, z) + 4|a_{1r}|^2 I_{\text{sw}}(\theta, z) \cos^2(2\pi x/p) \\ &\quad + 4|a_{0r}a_{1r}| e^{-\alpha_{\text{eff}} z} \cos(2\pi n_2 z(1 - \cos\theta)/\lambda) \cos(2\pi x/p) \end{aligned} \quad (4.122)$$

where $I_{\text{sw}}(\theta, z) = |E_{\text{sw}}(\theta, z)|^2$ and $\alpha_{\text{eff}} = \frac{\alpha}{2} \left(1 + \frac{1}{\cos\theta} \right)$.

Each diffraction order (plane wave) produces a standing wave. The interaction between the zero and first orders produces the interference pattern that forms the image, but the difference in propagation angles also creates a defocus term that accounts for defocusing of the image through the resist thickness. The terms that were ignored in this expression describe the defocusing of the reflected waves.

4.7.3 Resist-Induced Spherical Aberrations

Even with TE-polarized light and no substrate reflection, the resist causes important changes to the nature of the image. Consider Equation (4.122), three-beam imaging for TE illumination, for the case of no substrate reflection:

$$I(x, z) = |a_{0r}|^2 e^{-\alpha z} + 4|a_{1r}|^2 e^{-\alpha z / \cos \theta} \cos^2(2\pi x/p) + 4|a_{0r}a_{1r}| e^{-\alpha_{\text{eff}} z} \cos(2\pi n_2 \delta(1 - \cos \theta)/\lambda) \cos(2\pi x/p) \quad (4.123)$$

where $\delta = z - z_0$, and z_0 is the position of the plane of best focus. As a comparison, consider the same case for an aerial image:

$$I_{\text{air}}(x, z) = |a_0|^2 + 4|a_1|^2 \cos^2(2\pi x/p) + 4|a_0a_1| \cos(2\pi n_1 \delta(1 - \cos \theta)/\lambda) \cos(2\pi x/p) \quad (4.124)$$

There are three important and distinct differences between the image in air and the image in resist. First, as given in Equation (4.120), the amplitude of each diffraction order is different in the resist compared to air, and for the case of the first order this difference is angle, and thus pitch, dependent. Second, each diffracted order is absorbed differently also according to its angle, and thus pitch. In fact, an effective z -dependent diffraction order can be defined as

$$a_{0rz} \equiv a_{0r} e^{-\alpha z/2}, \quad a_{1rz} \equiv a_{1r} e^{-\alpha z/(2 \cos \theta)} \quad (4.125)$$

so that the image in resist becomes

$$I(x, z) = |a_{0rz}|^2 + 4|a_{1rz}|^2 \cos^2(2\pi x/p) + 4|a_{0rz}a_{1rz}| \cos(2\pi n_2 \delta(1 - \cos \theta)/\lambda) \cos(2\pi x/p) \quad (4.126)$$

As the image propagates further into the resist, the first orders are attenuated due to absorption to a greater degree than the zero order. This difference is more acute for larger angles (smaller pitches).

A more subtle but important difference between the images in air and resist is the defocus term, which includes the normalized optical path difference term $n\delta(1 - \cos \theta)/\lambda$. Both the refractive index and the angle θ are different in resist than in air. The results are depicted graphically in Figure 4.33, where larger angles (corresponding to smaller pitches) are focused further into the resist than smaller angles. First, the plane of best focus is shifted due to refraction. Further, there is a pitch dependence to best focus, a characteristic of spherical aberration.

To compare the defocus term in air to the case in resist, we can express the $\cos \theta$ as a sine and expand the resulting square root in a Taylor series.

$$OPD = n\delta(1 - \cos \theta) = n\delta(1 - \sqrt{1 - \sin^2 \theta}) = \frac{1}{2} n\delta \left(\sin^2 \theta + \frac{\sin^4 \theta}{4} + \dots \right) \quad (4.127)$$

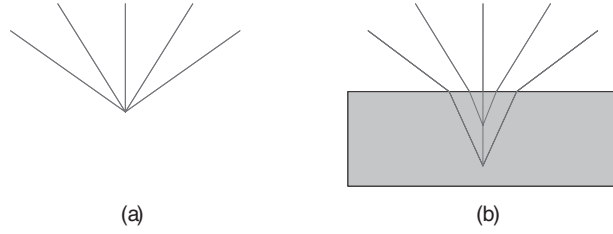


Figure 4.33 Focusing of plane waves arriving at different angles (a) in air, and (b) in resist, showing that the resist induces spherical aberration

Using Snell's law to compare the angles and refractive indices in air and resist,

$$\begin{aligned}
 OPD_{\text{air}} &= \frac{1}{2} n_1 \delta \left(\sin^2 \theta_1 + \frac{\sin^4 \theta_1}{4} + \dots \right) \\
 OPD_{\text{resist}} &= \frac{1}{2} n_2 \delta \left(\sin^2 \theta_2 + \frac{\sin^4 \theta_2}{4} + \dots \right) = \frac{1}{2} \frac{n_1}{n_2} n_1 \delta \left(\sin^2 \theta_1 + \frac{n_2^2}{n_1^2} \frac{\sin^4 \theta_2}{4} + \dots \right)
 \end{aligned} \tag{4.128}$$

and

$$\frac{OPD_{\text{resist}}}{OPD_{\text{air}}} = \frac{n_1}{n_2} \frac{\left(1 + \frac{\sin^2 \theta_2}{4} + \dots \right)}{\left(1 + \frac{\sin^2 \theta_1}{4} + \dots \right)} \tag{4.129}$$

In the paraxial limit, where only the lowest order term in the series above is kept, the impact of defocus is reduced in resist compared to that in air by the factor n_1/n_2 . The distance of the plane of best focus relative to the top of the resist is scaled by this factor. Also, the defocusing of the image through the resist is scaled by this amount. Since $n_2 > n_1$, the plane of best focus is shifted down into the resist, and the thickness of resist causes less defocusing than when the image propagates through the same thickness of air.

To see how the resist induces spherical aberration, we need only keep one more term in the series of Equation (4.129).

$$OPD_{\text{resist}} \approx \frac{n_1}{n_2} OPD_{\text{air}} - \delta \left[\frac{n_1^2 (n_2^2 - n_1^2)}{8 n_2^3} \right] \sin^4 \theta_1 \tag{4.130}$$

The added term varies as the radial position in the pupil to the fourth power, characteristic of 3rd order spherical aberration. Comparing to this Zernike term as given in Chapter 3, an effective amount of 3rd order spherical induced by the resist would be

$$Z_8 \approx -\frac{z}{\lambda} \left[\frac{n_1^2 (n_2^2 - n_1^2)}{24 n_2^3} \right] NA^4 \tag{4.131}$$

Consider a typical case where $n_1 = 1$ and $n_2 = 1.7$:

$$Z_8 \approx -0.016 \frac{z}{\lambda} NA^4 \quad (4.132)$$

For an NA of 0.9, 10 milliwaves of spherical aberration is induced per wavelength of resist thickness. Since resists tend to be between one and two wavelengths thick for such a high NA, the center of the resist can be expected to show between 5 and 10 milliwaves of effective spherical aberration. Since this resist-induced spherical aberration is systematic, it is possible to correct for it by adding an equivalent amount of oppositely directed spherical aberration into the lens design.

4.7.4 Standing Wave Amplitude Ratio

Our goal is now to understand how reflectivity, as well as the variation of reflectivity with angle, affects standing waves in the image. To begin, we'll define a metric called the standing wave amplitude ratio (SWAR), given by

$$SWAR = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \approx \frac{I_{\max} - I_{\min}}{2I(\rho = 0)} \quad (4.133)$$

Here, $I(\rho = 0)$ denotes the intensity that would be present if there were no reflecting substrate and is approximately the average of the min and max intensities. For a single plane wave, such as Equation (4.31), the SWAR is equal to twice the substrate amplitude reflection coefficient $\rho(\theta)$, and would be zero if the BARC were perfect. In fact, one goal of BARC design is to make the SWAR as close to zero as possible.

For the imaging case, pulling the maximum and minimum values out of Equation (4.118) or Equation (4.122) is complicated by two factors: the defocus dependence of the image, and the fact that each diffraction order produces a standing wave of a different period. If, however, the variation of the image intensity with z caused by defocus is small compared to the standing wave variation (i.e. if the depth of focus is much larger than one standing wave half-period), and best focus is near the position in the resist where the SWAR is being evaluated, this effect can be ignored. Further, by looking at the standing waves at the bottom of the resist, the mismatch in standing wave peak positions will be at their minimum. Under these circumstances, an approximate value for the SWAR can be obtained for this three-beam imaging case. For the simplified case ignoring absorption and multiple reflections in the resist,

$$SWAR \approx 2 \frac{a_o^2 |\rho(0)| + 4a_o a_1 \cos(2\pi x/p) \left(\frac{|\rho(0)| + |\rho(\theta)|}{2} \right) + 4a_1^2 \cos^2(2\pi x/p) |\rho(\theta)|}{a_o^2 + 4a_o a_1 \cos(2\pi x/p) + 4a_1^2 \cos^2(2\pi x/p)} \quad (4.134)$$

It is very interesting to note that the standing wave amplitude ratio is a function of x . Consider $x = 0$, the center of the space of the line/space pattern. Here, the SWAR is an average of the zero- and first-order reflectivities, weighted in a certain way by the amplitudes of the diffracted orders.

$$SWAR(x = 0) \approx 2 \frac{|\rho(0)|(a_o^2 + 2a_o a_1) + |\rho(\theta)|(4a_1^2 + 2a_o a_1)}{a_o^2 + 4a_o a_1 + 4a_1^2} \quad (4.135)$$

At $x = p/4$, corresponding to the nominal edge of the feature, the SWAR takes on a very different value.

$$\text{SWAR}(x = p/4) \approx 2|\rho(0)| \quad (4.136)$$

In other words, the standing waves at the edge of the feature are controlled only by the reflectivity of the zero order. Figure 4.34 shows an example of how the standing wave amplitude ratio varies with position along the line/space features. Note that Equations (4.134)–(4.136) and Figure 4.34 all assume a nonabsorbing medium. A real resist, with its reasonably high absorption, will reduce the actual SWAR significantly. Equation (4.134) can be modified to account for absorption and multiple reflections in a real resist if the diffraction order amplitudes a_0 and a_1 are replaced by their in-resist values a_{0r} and a_{1r} , and the substrate reflectivity ρ is replaced by

$$\rho(\theta)\tau_D^2(\theta)e^{-\alpha z/\cos\theta} \quad (4.137)$$

While the mathematics derived above apply to a fairly simple case, the results are appropriate for considering the approach toward optimizing a bottom antireflection coating for three-beam, high-numerical-aperture imaging (for two-beam imaging there is no ambiguity, since there is only one angle to optimize for). Equation (4.135) shows that standing waves in the middle of the space are controlled by a weighted average of the reflectivities of the zero and first orders (i.e. at zero angle and at the angle of the first orders). However, when considering the more important standing waves at the edge of the feature, only the zero-order reflectivity matters. While Equation (4.136) applies to both s - and p -polarized illumination, Equation (4.135) can be modified to approximate p -polarization by replacing a_1 with $a_1\cos\theta$ (to account for reduced interference creating the image) and by adding a $\cos 2\theta$ term to the weighting of the high-angle reflectivity (to account for reduced interference in the standing wave formation). Thus, for p -polarization,

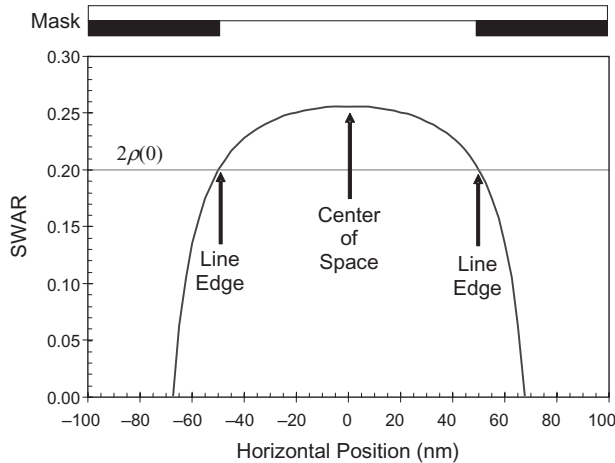


Figure 4.34 The standing wave amplitude ratio (SWAR) at different positions on the feature for coherent three-beam imaging and s -polarization. For this example of three-beam imaging of 100-nm lines and spaces, $a_0 = 0.5$, $a_1 = 0.3183$, $|\rho(0)| = 0.1$ and $|\rho(\theta)| = 0.15$

even the SWAR in the center of the space becomes more heavily weighted to the normally incident reflectivity due to the reduced interference of the large-angle first diffracted orders. It seems that designing a BARC by reducing the normally incident reflectivity provides a very good starting point for obtaining the best standing wave control. Beyond this initial design, full lithographic simulation will be required.

4.8 Defining Intensity

While seemingly simple in concept, the definition of light intensity is more complicated than expected. In particular, a comparison of intensity values when the light is in different materials and traveling at different angles requires careful consideration. One case where these difficulties become apparent is the simple refraction of a plane wave traveling from one medium to another. Thus, our discussion will begin with a look at electric field and intensity reflection and transmission coefficients. The following derivations are based on the standard treatment given in Born and Wolf.¹²

4.8.1 Intensity at Oblique Incidence

When determining the intensity (also called irradiance) transmitted into a material at an oblique angle, it is very important to understand the exact definition of intensity. By definition (and as was discussed briefly in Chapter 2), the *intensity* of light is the magnitude of the (time averaged) Poynting vector, the energy per second crossing a unit area perpendicular to the direction of propagation of the light. In a medium of refractive index n the intensity I is given by

$$I = n|E|^2 \quad (4.138)$$

where E is the electric field. Note that the definition given in Equation (4.138) may differ by a constant multiplicative factor depending on the units used.

The intensity reflectivity and transmission, for either polarization, are derived by considering a unit area on the interface between two materials labeled 1 and 2. Consider the projected power per unit area, J , of the incident light along the surface of the interface between the materials. If the angle of incidence is θ_1 ,

$$J_i = I_i \cos(\theta_1) \quad (4.139)$$

Likewise, the projected power per unit area of the reflected and transmitted light along this surface are

$$\begin{aligned} J_r &= I_r \cos(\theta_1) \\ J_t &= I_t \cos(\theta_2) \end{aligned} \quad (4.140)$$

Now the reflectivity and transmission coefficients for each polarization can be defined:

$$\begin{aligned} R_{12} = R_{21} = R &= \frac{J_r}{J_i} = |\rho_{12}|^2 \\ T_{12} = T_{21} = T &= \frac{J_t}{J_i} = \left| \frac{n_2 \cos(\theta_2)}{n_1 \cos(\theta_1)} \right| |\tau_{12}|^2 \end{aligned} \quad (4.141)$$

From these two equations it is easy to show that $R + T = 1$ for each polarization, which is a consequence of conservation of energy. Figure 4.5 shows how the intensity reflectivity varies with incident angle for both s - and p -polarized illumination for a typical air–resist interface. An alternate form for Equation (4.141), making use of the reverse direction definitions of reflection and transmission coefficients in Equation (4.38), are

$$\begin{aligned} R &= |\rho_{12}\rho_{21}| \\ T &= |\tau_{12}\tau_{21}| \end{aligned} \quad (4.142)$$

Also note that for randomly polarized light, the overall intensity reflectivity and transmittance can be computed from the values obtained for each polarization:

$$\begin{aligned} R &= R_{\parallel} \cos^2 \theta_i + R_{\perp} \sin^2 \theta_i \\ T &= T_{\parallel} \cos^2 \theta_i + T_{\perp} \sin^2 \theta_i \end{aligned} \quad (4.143)$$

Consider a unit intensity plane wave incident on the plane boundary between materials 1 and 2 at an incident angle θ_i and with intensity I_i . From Equation (4.138), the magnitude of the incident electric field must be

$$|E_i| = \sqrt{\frac{I_i}{n_1}} \quad (4.144)$$

The transmitted electric field is then

$$|E_t| = |\tau_{12}E_i| = |\tau_{12}| \sqrt{\frac{I_i}{n_1}} \quad (4.145)$$

The transmitted intensity (i.e. the intensity in material 2) is found by applying the definition of intensity to Equation (4.145).

$$I_t = n_2 |E_t|^2 = \frac{n_2}{n_1} |\tau_{12}|^2 I_i \quad (4.146)$$

By comparing Equation (4.146) with Equation (4.141), the somewhat nonintuitive result below is obtained.

$$\frac{I_t}{I_i} = T \frac{\cos(\theta_1)}{\cos(\theta_2)} \quad (4.147)$$

As can be seen in Equation (4.147), the transmittance T is not the ratio of the intensities I_t and I_i (see Figure 4.35). The difference comes from the change in the direction of the energy flow caused by refraction. Thus, one might ask the question, which is more important to know inside film 2, the intensity of the plane wave, or its projected power along a surface parallel to the material interface? The answer to this question depends on the task at hand, as discussed in section 4.8.3.

4.8.2 Refraction into an Absorbing Material

The well-known Snell's law of refraction [Equation (4.30)] dictates how a plane wave, traveling across a plane boundary between two materials of different optical properties,

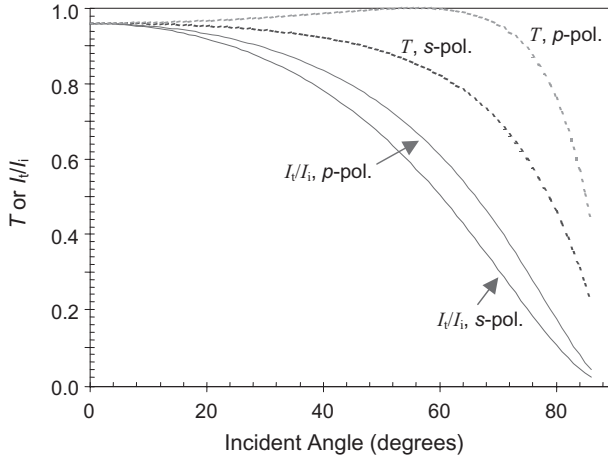


Figure 4.35 Intensity transmitted into layer 2 relative to the incident intensity (solid lines) and the transmittance T (dashed lines) as a function of the angle of incidence for both s- and p-polarization ($n_1 = 1.0$, $n_2 = 1.5$)

will change its direction. The material boundary conditions required by Maxwell's equations state that any electric field tangential to the material interface must be continuous across that interface. This requirement will lead directly to the derivation of Snell's law. When both refractive indices are real (i.e. neither material is absorbing), the interpretation of Snell's law is quite straightforward. However, if one or both of the materials are absorbing (for example, light traveling from air into photoresist), one or both of the angles will be complex. How is a complex angle of transmittance to be interpreted? To answer this question, a more rigorous treatment of refraction and absorption is required.

Assuming a unit amplitude plane wave, the electric field can be expressed mathematically in two dimensions as

$$E_i(x, z) = \exp(i2\pi n_1(x \sin \theta_1 + z \cos \theta_1)/\lambda) \quad (4.148)$$

where the interface between the two materials lies in the $x - y$ plane, positive x is up and positive z is to the right (as seen in Figure 4.4). Here we shall assume for simplicity that material 1 is nonabsorbing. The transmitted electric field will be

$$E_t(x, z) = \tau_{12} \exp(i2\pi n_2(x \sin \theta_2 + z \cos \theta_2)/\lambda) \quad (4.149)$$

where the appropriate expression for τ_{12} is used depending on the polarization of the incident plane wave. If material 2 is absorbing, the angle θ_2 will be complex. The resulting mathematics are quite cumbersome when trying to express all quantities as real numbers. For a weakly absorbing film, however, some simplifications are possible. The optical distance in Equation (4.149) is

$$OD = n_2(x \sin \theta_2 + z \cos \theta_2) \quad (4.150)$$

Applying Snell's law to eliminate the complex angle θ_2 ,

$$n_2 \cos \theta_2 = n_2 \sqrt{1 - \sin^2 \theta_2} = \sqrt{n_2^2 - (n_1 \sin \theta_1)^2} \quad (4.151)$$

If material 2 is weakly absorbing, the imaginary part of n_2 will be much smaller than the real part (for example, a highly absorbing photoresist may have $n_2 = n_2 + i\kappa_2 = 1.7 + i0.02$). Thus,

$$n_2^2 = n_2^2 - \kappa_2^2 + i2n_2\kappa_2 \approx n_2^2 + i2n_2\kappa_2 \quad (4.152)$$

where the error in neglecting the κ_2^2 term is only about 0.01 % for the example photoresist. Thus,

$$n_2 \cos \theta_2 \approx n_2 \sqrt{1 - \left(\frac{n_1 \sin \theta_1}{n_2} \right)^2} + i \frac{2\kappa_2}{n_2} \quad (4.153)$$

Now let us define a new angle called θ'_2 given by

$$\sin(\theta'_2) = \frac{n_1 \sin(\theta_1)}{n_2} \quad (4.154)$$

This new angle is not the actual transmitted angle, but is the angle of transmission that would be obtained if material 2 were nonabsorbing. For a weakly absorbing material, it is approximately the angle of transmission. Using this angle in Equation (4.153),

$$n_2 \cos \theta_2 \approx n_2 \sqrt{\cos^2 \theta'_2 + i \frac{2\kappa_2}{n_2}} = n_2 \cos \theta'_2 \sqrt{1 + i \frac{2\kappa_2}{n_2 \cos^2 \theta'_2}} \quad (4.155)$$

Expanding the square root into a Taylor series,

$$\sqrt{1 + i \frac{2\kappa_2}{n_2 \cos^2 \theta'_2}} = 1 + i \frac{\kappa_2}{n_2 \cos^2 \theta'_2} + \frac{1}{2} \left(\frac{\kappa_2}{n_2 \cos^2 \theta'_2} \right)^2 - \frac{i}{2} \left(\frac{\kappa_2}{n_2 \cos^2 \theta'_2} \right)^3 + \dots \quad (4.156)$$

If only the first two terms of the Taylor series are kept, both the real and imaginary parts of the result will have a relative error of approximately $\frac{1}{2} \left(\frac{\kappa_2}{n_2 \cos^2 \theta'_2} \right)^2$. Using again the optical properties of photoresist mentioned above and assuming a maximum incident angle of 90° in air, neglecting this term will provide less than 0.02 % error. Even if extreme immersion lithography is considered, the larger angle inside the resist will cause this error to grow to only 0.04 %. Thus, there is very little error in dropping all but the first two terms in Equation (4.156). The final result for Equation (4.155) is

$$n_2 \cos \theta_2 \approx n_2 \cos \theta'_2 + i \frac{\kappa_2}{\cos \theta'_2} \quad (4.157)$$

Thus, the plane wave traveling in material 2 will have an optical path of approximately

$$OD \approx xn_1 \sin \theta_1 + z \left(n_2 \cos \theta'_2 + i \frac{\kappa_2}{\cos \theta'_2} \right) = n_2(x \sin \theta'_2 + z \cos \theta'_2) + i \frac{z\kappa_2}{\cos \theta'_2} \quad (4.158)$$

and the electric field will be

$$E_t(x, y, z) \approx \tau_{12} \exp(i2\pi n_2(x \sin \theta'_2 + z \cos \theta'_2)/\lambda) \exp(-2\pi\kappa_2(z/\cos \theta'_2)/\lambda) \quad (4.159)$$

For a weakly absorbing film, the phase term of the electric field is approximately equal to the plane wave that would result for a nonabsorbing film with the same real part of the refractive index. The absorption term in Equation (4.159) provides the somewhat intuitive result that larger angles of propagation with respect to the z -axis give faster attenuation of the light along the z -axis. Because the surfaces of constant phase are not parallel to the surfaces of constant amplitude (which instead are parallel to the interface between the two materials), such a wave is called an *inhomogeneous wave*.

Applying the definition of intensity,

$$I_t(x, y, z) \approx n_2 |\tau_{12}|^2 \exp(-\alpha z / \cos \theta'_2) \quad (4.160)$$

where $\alpha = 4\pi\kappa/\lambda$ is the absorption coefficient. Note that the transmitted plane wave has an intensity with only z -dependence.

4.8.3 Intensity and Absorbed Energy

In lithography, the intensity of light inside the resist film is required in order to know the energy absorbed per unit volume in the resist, which in turn determines the amount of chemical change that occurs during exposure. The standard approach to photoresist exposure kinetics makes use of the Dill equation (to be discussed in more detail in Chapter 5),

$$\frac{dm}{dt} = -CI m \quad (4.161)$$

where m is the relative concentration of photosensitive material, C is the exposure rate constant, and t is the exposure time. Obviously, Equation (4.161) makes use of a specific definition of intensity. What definition is used, and why?

A detailed study of the microscopic mechanism of absorption and exposure can be used to relate the exposure rate constant C to the molar absorptivity of the photosensitive component of the resist (see Chapter 5):

$$C = \Phi a_m \lambda N_A h c \quad (4.162)$$

where Φ is the quantum efficiency, a_m is the molar absorptivity of the photosensitive component, λ is the vacuum wavelength, N_A is Avogadro's number, h is Planck's constant, and c is the speed of light. The important aspect of this relationship is the direct proportionality between the exposure rate constant and the molar absorptivity of the photosensitizer (which is the absorption coefficient divided by the concentration of photosensitizer). Thus, the rate of exposure in Equation (4.161) is directly proportional to the absorption coefficient times the intensity of light. However, this same rate of exposure is also directly proportional to the absorbed energy per unit volume of resist per second. After all, it is the absorbed energy that actually causes the chemical reactions to occur. Thus, our use of the intensity of light in the exposure rate Equation (4.161) must be based on the idea that

$$\text{Absorbed Energy/volume/time} \propto \alpha I \quad (4.163)$$

Our definition of intensity, for it to be useful in this application, must satisfy Equation (4.163). In fact, this can only be true if intensity is defined as the power per unit area perpendicular to the direction of travel. Consider first Lambert's law of absorption:

$$\frac{dI}{ds} = -\alpha I \quad (4.164)$$

where s is the direction of propagation of light. The change in intensity with s is just the absorbed power per unit volume when I is defined as power per unit area perpendicular to the direction of propagation. Further tying this definition of intensity together with the standard formulations for electric field propagation, absorption is accounted for by the use of a complex refractive index. For a homogeneous film, the solution to Lambert's law is

$$I(x) = I_0 e^{-\alpha s} \quad (4.165)$$

In phasor representation, a plane wave traveling in the s direction is written as

$$E(s) = A e^{i2\pi ns/\lambda} \quad (4.166)$$

Equation (4.166) will yield Equation (4.165) by applying the definition of intensity as given in Equation (4.138) if the absorption coefficient is related to the imaginary part of the refractive index by

$$\alpha = 4\pi\kappa/\lambda \quad (4.167)$$

If the direction s makes an angle θ with respect to the z -axis, then $s = z/\cos\theta$ and

$$I(z) = n|A|^2 \exp(-\alpha z/\cos\theta) = I_0 \exp(-\alpha z/\cos\theta) \quad (4.168)$$

Thus, one can see that by defining intensity as the energy flux perpendicular to the direction of travel of the light, the use of an imaginary refractive index in the standard optics equations provides for a straightforward calculation of energy absorbed per unit volume of resist. It is the intensity within the resist film, not J , the intensity projected along a surface parallel to the resist and substrate, that determines the amount of chemical change during exposure.

Problems

- 4.1. Imagine two plane waves moving in opposite directions along the z -axis. Using the time-dependent form of the waves rather than the phasor representation, the $+z$ moving E-field wave is given by

$$U_+(z, t) = A \cos(\omega t - kz)$$

and the $-z$ moving wave is

$$U_-(z, t) = A \cos(\omega t + kz)$$

Show that the sum of these waves is separable in z and t (that is, can be expressed as product of a function of z and a function of t). Show that this results in an intensity 'standing wave', that is, the time average of the square of the electric field varies sinusoidally with position.

- 4.2. Which of the following situations will result in the formation of standing waves? Explain your reasoning.
- (a) the superposition of identical waves that travel in the same direction;
 - (b) the superposition of identical waves that travel in opposite directions;
 - (c) the superposition of waves that are nearly identical but of different amplitudes that travel in opposite directions;
 - (d) the superposition of nearly identical waves of slightly different frequencies that travel in the same direction.
- 4.3. Derive Equation (4.19).
- 4.4. The *optical admittance* of a material, η , at normal incidence is just its refractive index, n . For light traveling through that material at an angle θ , the optical admittance of the material for TE (*s*-polarized) and TM (*p*-polarized) waves are

$$\eta_{\text{TE}} = n \cos \theta \quad \eta_{\text{TM}} = \frac{n}{\cos \theta}$$

Show that by using optical admittance in the Fresnel formulas [Equation (4.29)], one obtains a single equation for reflection for both polarizations that has the same form as the normally incident definition of Equation (4.8). What are the results for transmittance and how do they compare to Equation (4.7)?

- 4.5. *Brewster's angle* is defined as the angle at which *p*-polarized light will have zero reflectance at the interface between two materials. Derive an equation for Brewster's angle. For *p*-polarized light incident on a photoresist-coated substrate at Brewster's angle, what is the resulting swing amplitude?
- 4.6. For *p*-polarized light traveling at 45° with respect to the normal inside of a photoresist, no standing waves will occur since the incident and reflected waves will have no overlap in their electric fields [see Equation (4.37)]. Will the resist swing curve similarly have zero amplitude at this angle? What if the light is traveling at 45° with respect to the normal in air? Explain.
- 4.7. Derive Equation (4.46) from Equation (4.45).
- 4.8. Derive an expression for the CD swing curve using Equations (4.51)–(4.53).
- 4.9. Derive Equation (4.84).
- 4.10. Calculate the swing ratio for a nominally 400-nm-thick 193-nm resist (assume typical properties) on a substrate of copper. What is the impact on this swing ratio if the maximum is at a greater thickness than the minimum, compared to the minimum at a greater thickness than the maximum?
- 4.11. Using the normal-incidence BARC solution given in Figure 4.16 for a 40-nm BARC thickness, what BARC thickness control is required to keep the BARC intensity reflectivity below 0.5 %?
- 4.12. At 248 nm, a typical resist on silicon will result in a swing amplitude of about 0.69. Using a BARC with a reflectivity of 1 %, the swing amplitude is reduced to about 0.095. Ignoring absorption, what level of resist thickness control is required for each of these two cases (resist on silicon and resist on a 1 % reflective BARC) to keep the effective dose error below 1 %.
- 4.13. Derive Equation (4.103), the refractive index of an ideal TARC.
- 4.14. As an example of the use of Equation (4.100), consider a 193-nm resist with $n_2 = 1.7$, $\alpha = 1.2 \mu\text{m}^{-1}$, $D = 200 \text{ nm}$ and $|\rho_{12}| = 0.26$. Suppose that 2 % dose error

due to resist thickness variations was considered acceptable. Generate a plot of maximum allowed substrate reflectivity ($|\rho_{23}|^2$) as a function of the resist thickness variation Δ .

- 4.15. What will be the impact of using immersion lithography (versus dry lithography at the same numerical aperture) in terms of the contrast of the image-in-resist for unpolarized illumination? Explain.
- 4.16. Derive Equation (4.117) from Equation (4.116).
- 4.17. For light incident on a plane interface between two materials, use Equation (4.141) to show that $R + T = 1$ for each polarization.

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